AUTHOR QUERIES

AUTHOR PLEASE ANSWER ALL QUERIES

PLEASE NOTE: We cannot accept new source files as corrections for your article. If possible, please annotate the PDF proof we have sent you with your corrections and upload it via the Author Gateway. Alternatively, you may send us your corrections in list format. You may also upload revised graphics via the Author Gateway.

Carefully check the page proofs (and coordinate with all authors); additional changes or updates WILL NOT be accepted after the article is published online/print in its final form. Please check author names and affiliations, funding, as well as the overall article for any errors prior to sending in your author proof corrections. Your article has been peer reviewed, accepted as final, and sent in to IEEE. No text changes have been made to the main part of the article as dictated by the editorial level of service for your publication.

AQ:1 = Please confirm or add details for any funding or financial support for the research of this article.

AQ:2 = Please check the Associate Editor line in the first footnote for correctness.

AQ:3 = Please provide city name for Carnegie Mellon University.

AQ:4 = Please provide the publisher name and publisher location for Ref. [16].

Improving Wald's (Approximate) Sequential Probability Ratio Test by Avoiding Overshoot

Lasse Fischer and Aaditya Ramdas

***Abstract*— Wald's sequential probability ratio test (SPRT) is a cornerstone of sequential analysis. Based on desired type-I, II error levels α, β, it stops when the likelihood ratio crosses certain thresholds, guaranteeing optimality of the expected sample size. However, these thresholds are not closed form and the test is often applied with approximate thresholds $(1-\beta)/\alpha$ and $\beta/(1-\alpha)$ (approximate SPRT). When $\beta > 0$, this neither guarantees error control at α, β nor optimality. When $\beta = 0$ (power-one SPRT), this method is conservative and not optimal. The looseness in both cases is caused by *overshoot*: the test statistic overshoots the thresholds at the stopping time. Numerically calculating thresholds may be infeasible, and most software packages do not do this. We improve the approximate SPRT by modifying the test statistic to avoid overshoot. Our 'sequential boosting' technique *uniformly* improves power-one SPRTs ($\beta = 0$) for simple nulls and alternatives, or for one-sided nulls and alternatives in exponential families. When $\beta > 0$, our techniques provide guaranteed error control at α, β, while needing less samples than the approximate SPRT in our simulations. We also provide several nontrivial extensions: confidence sequences, sampling without replacement and conformal martingales.**

***Index Terms*— Anytime-valid, boosting, e-value, SPRT, test martingale, truncation.**

I. Introduction

To begin with a simple setting that we relax significantly later on, suppose we observe a stream of i.i.d. data $X_1, X_2, \ldots$ and are interested in a sequential test for

$$H_0 : X_i \sim \mathbb{P}_0 \quad \text{vs.} \quad H_1 : X_i \sim \mathbb{P}_1. \tag{1}$$

In particular, suppose we seek to control the type I and II error at level α, β for some predefined $\alpha, \beta \in [0, 1]$. The classical approach to this testing problem is the sequential probability ratio test (SPRT) introduced by Wand [1], [2]. Define the likelihood ratio

$$\Lambda_0 := 1 \quad \text{and} \quad \Lambda_t := \frac{\prod_{i=1}^t p_1(X_i)}{\prod_{i=1}^t p_0(X_i)} \ (t \in \mathbb{N}), \tag{2}$$

where p_0 and p_1 are the densities with respect to some measure μ of $\mathbb{P}_0$ and $\mathbb{P}_1$, respectively. For some chosen constants $0 \leq \gamma_0 < \gamma_1$, at each time t, Wald's SPRT does the following:

$$\begin{aligned} &\text{stop and reject } H_0 \text{ if } \Lambda_t \geq \gamma_1, \\ &\text{stop and accept } H_0 \text{ if } \Lambda_t \leq \gamma_0, \\ &\text{continue sampling if } \gamma_0 < \Lambda_t < \gamma_1. \end{aligned}$$

Let $\delta_{\gamma_0,\gamma_1} \in \{0, 1\}$ be the test decision of the SPRT (1 if H_0 is rejected and 0 if accepted) and τ_{γ_0,γ_1} the stopping time. Wand [3] showed that τ_{γ_0,γ_1} is almost surely finite. Furthermore, Wald and Wolfowitz [4] proved that for any other sequential test (δ', τ') it holds that

$$\begin{aligned} &\mathbb{P}_h(\delta' = 1 - h) \leq \mathbb{P}_h(\delta_{\gamma_0,\gamma_1} = 1 - h) \ (h \in \{0, 1\}) \\ \implies &\mathbb{E}_h(\tau') \geq \mathbb{E}_h(\tau_{\gamma_0,\gamma_1}) \ (h \in \{0, 1\}). \end{aligned} \tag{3}$$

Hence, the SPRT minimizes the expected sample size among all other sequential tests with the same or smaller type I and type II error. While this may seem to be the end of the story, the catch is that it is difficult to specify γ_0, γ_1 so that the error probabilities match desired levels. We expand on this below.

a) Wald's approximate SPRT: In practice, it may be quite difficult to calculate $\mathbb{P}_0(\delta_{\gamma_0,\gamma_1} = 1)$ and $\mathbb{P}_1(\delta_{\gamma_0,\gamma_1} = 0)$ exactly [1], [5]. Therefore, we cannot choose γ_0 and γ_1 such that the type I error is equal to a desired level $\alpha \in (0, 1)$ and the type II error is equal to a desired level $\beta \in (0, 1)$. For this reason, Wand [1], [2] introduced approximations for γ_1 and γ_0:

$$\gamma_1 = \frac{1-\beta}{\alpha} \text{ and } \gamma_0 = \frac{\beta}{1-\alpha}. \tag{4}$$

Wand [1] showed that at least one of the two errors will be controlled at the desired level:

$$\mathbb{P}_0(\delta_{\gamma_0,\gamma_1} = 1) + \mathbb{P}_1(\delta_{\gamma_0,\gamma_1} = 0) \leq \alpha + \beta. \tag{5}$$

Nevertheless, in many trials it is required that at least the type I error is controlled at the prespecified level α. Furthermore, even if the sum of type I and type II errors are controlled, it is not ensured that the SPRT with Wald's approximation minimize the expected sample size, meaning more efficient procedures are possible. The reason for the inaccuracy of Wald's approximation is that it is derived under

Received 11 June 2025; revised 16 January 2026; accepted 20 January 2026. The work of Lasse Fischer was supported by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Project 281474342/GRK2224/2. The work of Aaditya Ramdas was supported by NSF under Grant DMS-2310718. *(Corresponding author: Lasse Fischer.)*

Lasse Fischer is with the Competence Center for Clinical Trials Bremen, University of Bremen, 28359 Bremen, Germany (e-mail: fischer1@uni-bremen.de).

Aaditya Ramdas is with the Department of Statistics and Data Science, Carnegie Mellon University, PA 15213 USA (e-mail: aramdas@cmu.edu).

Communicated by S. Zou, Associate Editor for Machine Learning and Statistics.

This article has supplementary material provided by the authors and color versions of one or more figures available at https://doi.org/10.1109/TIT.2026.3658855.

D

© 2026 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

the assumption of no *overshoot*, meaning that at the stopping time τ_{γ_0,γ_1} the likelihood ratio equals exactly one of the boundaries γ_0 or γ_1.

To summarize, *the thresholds in (4) neither guarantee type I, II error control nor optimal use of sample size, and yet they are used very frequently in practice* (e.g. [6], [7], [8], [9]). In particular, the R packages "sprtt: Sequential Probability Ratio Tests Toolbox" [10] and "SPRT: Wald's Sequential Probability Ratio Test" [11] apply these approximate boundaries (the R Package "SPRT was archived and removed from CRAN in 2022)".

b) The conservative power-one SPRT: If we set $\beta = 0$ in Wald's approximated thresholds (4), then the SPRT stops for rejection if $\Lambda_t \geq 1/\alpha$, but never stops for accepting H_0. Interestingly, the SPRT provably controls the type I error at level α in this case. This follows, for example, by Ville's inequality which we will recap in Section II. In addition, it is still guaranteed that the SPRT stops almost surely after a finite number of samples if H_1 is true [3], and therefore is called conservative power-one SPRT, in particular studied and espoused by Robbins [12]. A level-α sequential test of power one (in short, a *power-one sequential test*) is defined by a stopping time τ (at which you reject the null) such that

$$\mathbb{P}_0(\tau < \infty) \leq \alpha \text{ and } \mathbb{P}_1(\tau < \infty) = 1, \tag{6}$$

where $\alpha \in (0,1)$ is a prespecified significance level. The power-one SPRT $\tau_{\gamma_1} := \inf\{t \in \mathbb{N} : \Lambda_t \geq \gamma_1\}$, $\gamma_1 \geq 1$, uses the available samples optimally in the following sense [13]:

$$\mathbb{P}_0(\tau' < \infty) \leq \mathbb{P}_0(\tau_{\gamma_1} < \infty) \implies \mathbb{E}_1(\tau') \geq \mathbb{E}_1(\tau_{\gamma_1}).$$

Consequently, any other sequential test that controls the type I error at the same level the SPRT does, needs on average at least as many samples as the SPRT to make a rejection if the alternative is true. Since it may be difficult to calculate $\mathbb{P}_0(\tau_{\gamma_1} < \infty)$ exactly, one often employs the conservative power-one SPRT by setting $\gamma_1 = 1/\alpha$ to ensure (6). But this only guarantees exact type I error control and thus optimality if $(\Lambda_t)_{t\in\mathbb{N}}$ never overshoots at γ_1 and almost surely attains 0 or γ_1 at some point [14]. However, this is usually not fulfilled.

c) Wald's conservative SPRT: The aforementioned Ville's inequality also implies that the type II error for Wald's two-sided SPRT is bounded by γ_0. Therefore, the SPRT with

$$\gamma_1 = 1/\alpha \text{ and } \gamma_0 = \beta$$

provides valid type I and type II error control at α and β respectively, which is why we refer to it as Wald's conservative SPRT in the following.

d) Outline of the paper and our contributions: In this paper, we propose a general *sequential boosting method* to uniformly improve (super)martingale based sequential tests by avoiding an overshoot at $1/\alpha$ (Section II). This means that our boosted sequential tests are never worse than the initial tests, but may stop earlier and are more likely to achieve a rejection. In Section III, we use this to uniformly improve conservative power-one SPRTs for simple and one-sided nulls and alternatives. We provide nontrivial extensions and applications of our method in Section IV, improving some existing confidence sequences, methods for sampling without replacement and conformal martingales. In Section V, we demonstrate how our boosting approach should be applied if a stop for futility is to be included and type II error control is desired. Finally, we discuss extensions of our method and existing approaches for dealing with overshoot when using the SPRT (Section VI). All formal proofs are provided in the supplementary material.

In Table I we compare Wald's SPRTs with our boosted versions. The type I error is denoted by α' and the type II error by β'. In the power-one case, our SPRT provides the same guarantees as Wald's SPRT, but uniformly improves it in terms of sample size. For the two-sided case, Wald's conservative SPRT and our boosted SPRT provide valid type I and type II error control while Wald's approximate SPRT does not. In addition, our two-sided SPRT outperformed Wald's approximate SPRT with respect to sample size in all considered simulation scenarios and provably improves Wald's conservative SPRT.

e) Comparison to existing alternatives of Wald's approximated thresholds: We do *not* claim that our boosted tests are optimal in the sense of (3). Hence, if the exact parameters γ_0 and γ_1 for obtaining a desired type I and type II error guarantee with the SPRT are known, one should employ the usual SPRT with these thresholds. For example, this can be the case in settings where the (asymptotic) approximations by Siegmund [5] are reliable (see Section VI for a more detailed discussion of this approach). However, we will see that our approach can be applied in a much broader range of settings, including in some nonparametric settings (like testing exchangeability using conformal martingales), and provides non-asymptotic control of the type I and type II error. Further advantages over Siegmund's and other approaches, such as the use of numerical methods to determine the exact thresholds, are discussed in Section VI-B.

II. Main Technique: Improving Test Martingales by Avoiding Overshoot

A process $(M_t)_{t\in\mathbb{N}_0}$ adapted to some filtration $(\mathcal{F}_t)_{t\in\mathbb{N}_0}$ (in the following we denote processes and filtrations just by (M_t) and $(\mathcal{F})$, respectively) is a test martingale [15] for $\mathbb{P}_0$ if (M_t) is nonnegative, $\mathbb{E}_0[M_0] = 1$ and $\mathbb{E}_0[M_t|\mathcal{F}_{t-1}] = M_{t-1}$. We will also speak of test supermartingale, if the last equality is replaced by an inequality. Ville's inequality [16] states that for any test supermartingale (M_t) it holds that

$$\mathbb{P}_0(\exists t \in \mathbb{N} : M_t \geq 1/\alpha) \leq \alpha \quad (\alpha \in (0,1)). \tag{7}$$

For example, the likelihood ratio process (Λ_t) (2) is a test martingale, since

$$\begin{aligned}\mathbb{E}_0\left[\Lambda_t|\Lambda_{t-1}\right] &= \Lambda_{t-1}\mathbb{E}_0\left[\frac{p_1(X_t)}{p_0(X_t)}\right] \\ &= \Lambda_{t-1}\int_{\mathcal{X}} \frac{p_1(x)}{p_0(x)} p_0(x)\mu(dx) = \Lambda_{t-1},\end{aligned} \tag{8}$$

where $\mathcal{X}$ is the support of X_t under $\mathbb{P}_0$ (one could set $\mathcal{F}_t = \sigma(X_1,\ldots,X_t)$, but Λ_t depends on $\mathcal{F}_{t-1}$ only through Λ_{t-1} which is why we simply condition on Λ_{t-1}). This proves the type I error control of the conservative power-one

TABLE I

COMPARISON OF THE GUARANTEES PROVIDED BY WALD'S APPROXIMATE AND CONSERVATIVE SPRTS, TO OUR BOOSTED SPRTS. THE DESIRED TYPE I AND TYPE II ERROR PROBABILITIES ARE DENOTED BY α AND β, WHILE THE ACTUALLY ACHIEVED PROBABILITIES ARE DENOTED BY α' AND β'

Sequential test	Error control	Sample size
Conservative power-one SPRT	$\alpha' \leqslant \alpha, \beta' = 0$	Not optimal
Our boosted power-one SPRT	$\alpha' \leqslant \alpha, \beta' = 0$	Not optimal, but never stops later than power-one SPRT
Wald's approximate SPRT	$\alpha' \leqslant \frac{\alpha}{1-\beta}, \beta' \leqslant \frac{\beta}{1-\alpha}$ $\alpha' + \beta' \leqslant \alpha + \beta$	Not optimal
Wald's conservative SPRT	$\alpha' \leqslant \alpha, \beta' \leqslant \beta$	Not optimal
Our boosted two-sided SPRT	$\alpha' \leqslant \alpha, \beta' \leqslant \beta$	Not optimal, but often stops earlier than approx. SPRT and never later than cons. SPRT

SPRT ($\gamma_1 = 1/\alpha$, $\gamma_0 = 0$). However, Ville's inequality holds with equality only if (M_t) never overshoots at $1/\alpha$ [14] and therefore this power-one SPRT usually does not exhaust the type I error. We now introduce a general approach to uniformly improve (super)martingale based sequential tests by avoiding overshoot.

A. Truncation and Boosting of Test Supermartingales

Suppose we have some nonnegative process (M_t) with $\mathbb{E}_0[M_0] = 1$ that is adapted to a filtration $(\mathcal{F}_t)$. For example, one can think of $\mathcal{F}_t = \sigma(X_1, \dots, X_t)$ as being generated by the observations, however, in some cases it is useful to coarsen the filtration or add randomization, as we shall later see with conformal martingales. It should be noted that in this section we do not make any assumption about the distribution of the data (in particular, $X_1, X_2, \dots$ do not need to be i.i.d.). We define $L_t = M_t/M_{t-1}$ with the convention $0/0 = 0$ as the individual multiplicative factors of (M_t), so that

$$M_t = M_{t-1} \cdot L_t = \prod_{i=1}^{t} L_i \quad (t \in \mathbb{N}).$$

Then (M_t) is a test supermartingale with respect to $(\mathcal{F}_t)$ iff

$$\mathbb{E}_0[L_t | \mathcal{F}_{t-1}] \leq 1 \quad \text{for all } t \in \mathbb{N}. \tag{9}$$

Hence, every test supermartingale can be decomposed into its individual factors. Furthermore, if factors $(L_t)_{t\in\mathbb{N}}$ adapted to $(\mathcal{F}_t)$ satisfying (9) are given, we can construct a new test supermartingale by $M_t = \prod_{i=1}^{t} L_i$. Defining the stopping time

$$\tau_M := \inf\{t \in \mathbb{N}_0 : M_t \geq 1/\alpha\}$$

at which we reject the null, Ville's inequality implies type I error control: $\mathbb{P}_0(\tau_M < \infty) \leq \alpha$. Mathematically, a stopping time τ is an integer-valued random variable such that $\{\tau \leq t\}$ is measurable with respect to $\mathcal{F}_t$ for all $t \in \mathbb{N}$. The optional stopping theorem implies that $\mathbb{E}_0[M_\tau] \leq 1$ for all stopping times τ.

Our approach is to increase the individual factors L_t of a given test supermartingale (M_t), yet avoiding an overshoot at $1/\alpha$. As a first step, we define the truncation function

$$T_\alpha(x; M) = \begin{cases} x, & \text{if } Mx \leq \frac{1}{\alpha} \\ \frac{1}{M\alpha}, & \text{if } Mx > \frac{1}{\alpha}. \end{cases} \tag{10}$$

Note the key property that $M \cdot T_\alpha(x; M) \leq 1/\alpha$ for all $x \in [0, \infty]$. Now define the process

$$M_t^{\text{trunc}} = M_{t-1}^{\text{trunc}} \cdot T_\alpha(L_t L_t; M_{t-1}^{\text{trunc}}) = \prod_{i=1}^{t} T_\alpha(L_i; M_{i-1}^{\text{trunc}}). \tag{11}$$

By construction, M_t^{trunc} never overshoots at $1/\alpha$. Now, define

$$\tau_{\text{trunc}} := \inf\{t \in \mathbb{N}_0 : M_t^{\text{trunc}} \geq 1/\alpha\}.$$

We claim that

$$\tau_{\text{trunc}} = \tau_M,$$

meaning the sequential test based on (M_t^{trunc}) makes the same decision and at the same time as the test based on (M_t). This is because for $t < \tau_M$, $M_t^{\text{trunc}} = M_t$, meaning that the truncation has no effect if the threshold is not reached. At the very last step, when M_t reaches (but may potentially exceed) $1/\alpha$, M_t^{trunc} simply equals $1/\alpha$, proving our claim.

However, since $T_\alpha(L_t; M_{t-1}^{\text{trunc}}) \leq L_t$, we can potentially increase the factors $T_\alpha(L_t; M_{t-1}^{\text{trunc}})$ and thus stop sooner while retaining valid error control. We propose to improve $T_\alpha(L_t; M_{t-1}^{\text{trunc}})$ by multiplying L_t with a "boosting factor" $b_t \geq 1$ (the larger, the better): at each step t, we calculate b_t as large as possible such that

$$\mathbb{E}_0[T_\alpha(b_t L_t; M_{t-1}^{\text{boost}}) | \mathcal{F}_{t-1}] \leq 1, \tag{12}$$

where $M_{t-1}^{\text{boost}} = \prod_{i=1}^{t-1} T_\alpha(b_i L_i; M_{i-1}^{\text{boost}})$.

Note that without the truncation function T_α in (12), it is usually not possible to find a boosting factor $b_t > 1$, since, e.g., likelihood ratios have an expected value of exactly one. However, the truncation often permits $b_t > 1$, even if $\mathbb{E}_0[L_t | \mathcal{F}_{t-1}] = 1$. Now, (M_t^{boost}) is our boosted process that we use for testing. This procedure is summarized in Algorithm 1. Note that if $M_{t-1}^{\text{boost}} = 1/\alpha$, we could choose $b_t = \infty$, so we just set $M_t^{\text{boost}} = 1/\alpha$ in this case.

Theorem 1: Let (M_t) be any test supermartingale and (M_t^{boost}) be the boosted process obtained by Algorithm 1, yielding the sequential tests $\tau_M = \inf\{t \in \mathbb{N} : M_t \geq 1/\alpha\}$ and $\tau_{\text{boost}} = \inf\{t \in \mathbb{N} : M_t^{\text{boost}} \geq 1/\alpha\}$, respectively. Then $M_t^{\text{boost}} \geq M_t$ for all $t < \tau_M$ and $\tau_{\text{boost}} \leq \tau_M$. Furthermore, (M_t^{boost}) is a test supermartingale. Therefore, under the null,

Algorithm 1 Improving Test Supermartingales by Avoiding Overshoot

Input: Test supermartingale (M_t) and individual significance level α.

Output: Boosted test supermartingale (M_t^{boost}).

$M_0^{\text{boost}} = 1$
for $t = 1, 2, \dots$ **do**
  Define $L_t := M_t / M_{t-1}$
  Choose $b_t \geq 1$ as large as possible such that $\mathbb{E}_0[T_\alpha(b_t L_t; M_{t-1}^{\text{boost}})|\mathcal{F}_{t-1}] \leq 1$
  $M_t^{\text{boost}} = M_{t-1}^{\text{boost}} T_\alpha(b_t L_t; M_{t-1}^{\text{boost}})$
end for
return (M_t^{boost})

$\mathbb{P}_0(\tau_{\text{boost}} < \infty) \leq \alpha$ and $\mathbb{E}_0[M_\tau^{\text{boost}}] \leq 1$ for all stopping times τ.

Theorem 1 shows that Algorithm 1 provides a universal improvement of test supermartingales if α is prespecified. Regardless of which test supermartingale (M_t) is put into Algorithm 1, the test based on the output process (M_t^{boost}) always stops sooner ($\tau_{\text{boost}} \leq \tau_M$). Further, if we stop before the null is rejected, the boosted test martingale provides more evidence against the null than the initial process ($M_t^{\text{boost}} \geq M_t$ for all $t < \tau_M$).

Remark 1: The individual factors L_t have sometimes been termed 'sequential e-values' in the recent literature [17], [18]. A similar, but non-sequential, truncation approach like (10) was proposed by Wang and Ramdas [19] in the context of multiple testing. They also mentioned that in general we can use any "boosting function" $\psi_t : [0, \infty] \to [0, \infty]$ such that

$$\mathbb{E}_0[T_t(\psi_t(L_t); M_{t-1}^{\text{boost}})|\mathcal{F}_{t-1}] \leq 1 \quad \text{for all } t \in \mathbb{N}. \tag{13}$$

However, in a different context, parallel work by Koning [20] proved that $\psi_t(x) = b_t x$, where b_t is chosen such that (12) is satisfied with an equality, is the log-optimal choice. Therefore, we will stick to this particular boosting factor for the rest of the paper.

Remark 2: (M_t^{boost}) is a test martingale if (12) holds with equality. Therefore, (M_t^{boost}) can be interpreted as a likelihood ratio between $\mathbb{Q}$ and $\mathbb{P}_0$, where $\mathbb{Q}$ is adaptively determined.

B. Existence and Calculation of Optimal Boosting Factors

One question is, of course, if there is a largest possible boosting factor b_t and, if so, how we can determine it. In the following proposition, we show that there always exists a b_t^* that satisfies (12) with an equality and that if we have found such a b_t^*, then there is no further b_t which leads to a larger boosted martingale factor.

Proposition 2: If $\mathbb{P}_0(L_t = 0|\mathcal{F}_{t-1}) = 0$ and $M_{t-1}^{\text{boost}} \in (0, \alpha^{-1})$, then there exists b_t^* s.t.

$$\mathbb{E}_0[T_\alpha(b_t^* L_t; M_{t-1}^{\text{boost}})|\mathcal{F}_{t-1}] = 1. \tag{14}$$

For every $b_t \geq 1$ satisfying (12), we have $T_\alpha(b_t^* L_t; M_{t-1}^{\text{boost}}) \geq T_\alpha(b_t L_t; M_{t-1}^{\text{boost}})$, $\mathbb{P}_0$-a.s.

Since $T_\alpha(x; M_{t-1}^{\text{boost}})$ is nondecreasing in x, it is numerically very easy to find b_t^*; start with $b_t = 1$ and slightly increase it until (14) is satisfied with the desired accuracy. This can be solved accurately with standard root solving algorithms (we use the R function *uniroot* in our implementations). The only requirement is that we know the distribution of $L_t|\mathcal{F}_{t-1}$ under $\mathbb{P}_0$ such that we can calculate the expected value in (14) for arbitrary b_t. In the following section we provide a closed form for this expected value for SPRTs. In some cases (e.g., for binary data), it is even possible to obtain a closed form for b_t^* (see Section IV-B).

III. Improving Power-One SPRTs

In this section, we demonstrate how (conservative) power-one SPRTs can be improved by the boosting technique proposed in Algorithm 1. Hence, we consider the setting from the introduction and assume that $X_1, X_2, \dots$ are i.i.d. data. SPRTs are particularly well-suited for our method, since each individual factor L_t has the simple form $L_t = \lambda(X_t)$, where λ is the likelihood ratio between the null and the alternative. In addition, the distribution of $\lambda(X_t)$ under the null hypothesis is usually known.

Although only required in Subsection III-C, to unify the notation we assume that the distribution class under consideration can be parametrized by some parameter space Θ. The true but unknown parameter of interest will be denoted by $\theta^* \in \Theta$.

A. Simple Null Vs. Simple Alternative

In this subsection, we consider the simple testing problem

$$H_0 : \theta^* = \theta_0 \quad \text{vs.} \quad H_1 : \theta^* = \theta_1,\ \theta_1 \neq \theta_0. \tag{15}$$

In this case, the likelihood ratio is given by $\lambda(X_t) = \frac{p_{\theta_1}(X_t)}{p_{\theta_0}(X_t)}$. This allows to write the expected value in (12) in the following simple form

$$\begin{aligned}
&\mathbb{E}_{\theta_0}[T_\alpha(b_t\lambda(X_t); \Lambda_{t-1}^{\text{boost}})|\Lambda_{t-1}^{\text{boost}}] \\
&= \mathbb{E}_{\theta_0}\left[b_t\lambda(X_t)\mathbb{1}\left\{b_t\lambda(X_t) \leq \frac{1}{\alpha\Lambda_{t-1}^{\text{boost}}}\right\}\middle|\Lambda_{t-1}^{\text{boost}}\right] \\
&+ \mathbb{E}_{\theta_0}\left[\frac{1}{\Lambda_{t-1}^{\text{boost}}\alpha}\mathbb{1}\left\{b_t\lambda(X_t) > \frac{1}{\alpha\Lambda_{t-1}^{\text{boost}}}\right\}\middle|\Lambda_{t-1}^{\text{boost}}\right] \\
&= b_t\int_{\mathcal{X}} \frac{p_{\theta_1}(x)}{p_{\theta_0}(x)} p_{\theta_0}(x)\mathbb{1}\left\{b_t\lambda(x) \leq \frac{1}{\alpha\Lambda_{t-1}^{\text{boost}}}\right\}\mu(dx) \\
&+ \frac{1}{\Lambda_{t-1}^{\text{boost}}\alpha}\mathbb{P}_{\theta_0}\left(\lambda(X_t) > \frac{1}{b_t\alpha\Lambda_{t-1}^{\text{boost}}}\middle|\Lambda_{t-1}^{\text{boost}}\right) \\
&= b_t\mathbb{P}_{\theta_1}\left(\lambda(X_t) \leq \frac{1}{b_t\alpha\Lambda_{t-1}^{\text{boost}}}\middle|\Lambda_{t-1}^{\text{boost}}\right) \\
&+ \frac{1}{\Lambda_{t-1}^{\text{boost}}\alpha}\mathbb{P}_{\theta_0}\left(\lambda(X_t) > \frac{1}{b_t\alpha\Lambda_{t-1}^{\text{boost}}}\middle|\Lambda_{t-1}^{\text{boost}}\right),
\end{aligned} \tag{16}$$

where $\Lambda_{t-1}^{\text{boost}} = \prod_{i=1}^{t-1} T_\alpha(b_i\lambda(X_i); \Lambda_{i-1}^{\text{boost}})$. To calculate an optimal boosting factor, we can then just set the equation equal to 1 and solve for b_t numerically (see Proposition 2).

TABLE II

BOOSTING FACTORS OBTAINED FOR $\alpha = 0.05$ IN A SIMPLE GAUSSIAN TESTING PROBLEM WITH MEAN DIFFERENCE δ AND VARIANCE 1

δ \ $\Lambda_{t-1}^{\text{boost}}$	0.5	1	2	4	10
0.1	1	1	1	1	1.00001
0.5	1	1	1	1.00019	1.03019
1	1.00015	1.00157	1.01077	1.05386	1.37349
2	1.13931	1.27600	1.55046	2.17468	5.73972
3	2.45490	3.49439	5.72975	11.8255	68.1985

For example, in case of a simple Gaussian testing problem $H_0 : X_i \sim \mathcal{N}(\mu_0, 1)$ vs. $H_1 : X_i \sim \mathcal{N}(\mu_1, 1)$, $\mu_1 - \mu_0 = \delta > 0$, we obtain

$$\lambda(x) = \exp(\delta(x - \mu_0) - \delta^2/2) \text{ and } \lambda^{-1}(y) = \frac{\delta^2/2 + \log(y)}{\delta} + \mu_0.$$

Hence, (16) becomes

$$b_t \left[\Phi\left(\frac{\log\left([b_t \alpha \Lambda_{t-1}^{\text{boost}}]^{-1}\right)}{\delta} - \frac{\delta}{2}\right)\right] + (\Lambda_{t-1}^{\text{boost}}\alpha)^{-1}\left[1 - \Phi\left(\frac{\log([b_t \alpha \Lambda_{t-1}^{\text{boost}}]^{-1})}{\delta} + \frac{\delta}{2}\right)\right], \quad (17)$$

where Φ is the CDF of a standard Gaussian. In the following we give some specific boosting factors that were determined by setting (17) to one and solving for b_t numerically.

Suppose we test the above hypothesis for $\delta = 2$ at level $\alpha = 0.05$ and we are at the beginning of the testing process, meaning $\Lambda_{t-1}^{\text{boost}} = \Lambda_0^{\text{boost}} = 1$. Then we obtain a boosting factor of $b_t = 1.28$. If the current martingale value increases, the boosting factor increases as well, such that we obtain $b_t = 1.55$ for $\Lambda_{t-1}^{\text{boost}} = 2$, but $b_t = 1.14$ for $\Lambda_{t-1}^{\text{boost}} = 0.5$. In a similar manner, b_t is increasing in δ.

See Table II for boosting factors on a grid of values for δ and $\Lambda_{t-1}^{\text{boost}}$. If δ and $\Lambda_{t-1}^{\text{boost}}$ are both small, the boosting factors are close to one. The reason is that in this case the probability under H_0 of overshooting at $1/\alpha$ in the next step is close to zero. But also note that for small δ the expected sample size is usually large and this table only shows the boosting factors for one step. The total gain of boosting at some time could be calculated as the cumulative product of the boosting factors up to that time. The table also shows that the SPRT is very conservative for large δ. For example, if $\delta = 3$ and $\lambda(X_1) = \lambda(X_2) = 1$, then $\Lambda_2 = 1$, but we would already have $\Lambda_2^{\text{boost}} > 20$ such that the boosted test could already reject the hypothesis.

In Figure 1 we compare the SPRT with its boosted improvement regarding sample size and type I error in the simple Gaussian testing setup for different strengths of the signal δ, where $\mu_0 = 0$. The results were obtained by averaging over 10 000 trials. In each trial, we generated up to 10 000 observations under the alternative until the hypothesis could be rejected by the respective sequential test. Both tests rejected the hypothesis in all cases. In order to determine the type I error for both of the methods, we used importance sampling by exploiting the fact [21] that

$$\mathbb{P}_{\theta_0}(\tau < \infty) = \mathbb{E}_{\theta_1}\left[\mathbb{1}\{\tau < \infty\}\Lambda_\tau^{-1}\right], \quad (18)$$

where Λ_τ^{-1} is the likelihood ratio of θ_0 to θ_1 at time τ. With this, we can estimate the type I error by the inverse of the likelihood ratio at the stopping time while sampling under the alternative. Since τ is almost surely finite and usually much smaller under the alternative than under the null hypothesis, this reduces the required number of samples to obtain a good estimate of the type I error [5], [21].

The boosting approach saves on average 3.4% to 11.8% of the sample size compared to the SPRT in the considered scenarios, while the percentage increases with the strength of the signal. Furthermore, the boosted SPRT exhausts the significance level, while the SPRT is becoming increasingly conservative for stronger signals. For all considered δ, the mean computational time required for running one boosted SPRT, including the numerical calculation of the corresponding boosting factors, was below 0.05 seconds (performed on a laptop with an Intel Core i7-10510U CPU, 1.80GHz, 8 cores, 16 GB RAM), showing that the approach is computationally feasible in practice. In Supplementary Material S.2 we repeat this experiment for $\alpha = 0.01$ instead of $\alpha = 0.05$, which yields a similar result.

Remark 3: It is a well-known fact that in the simple hypothesis setting (15) the stopping time $\tau_{1/\alpha}$ of Wald's power-one SPRT satisfies $\lim_{\alpha \downarrow 0} \mathbb{E}_{\mathbb{Q}}[\tau_{1/\alpha}/\log(1/\alpha)] = 1/\text{KL}(\mathbb{Q}, \mathbb{P})$, where KL is the Kullback-Leibler divergence. Also, no other test can lead to a lower right-hand side. Since our boosted SPRT never stops later than Wald's SPRT, the boosted SPRT satisfies the same equality.

B. Handling Composite Alternatives With Predictable plugins

In practice, a concrete alternative may be hard to specify, leading to problems of the form

$$H_0 : \theta^* = \theta_0 \quad \text{vs.} \quad H_1 : \theta^* > \theta_0.$$

There are two approaches that are usually used to handle such composite alternatives in sequential testing: the method of mixtures [22] and the predictable plugin [1], [23], [24]. In the following, we will recap the latter, since it is better suited for our boosting technique.

Let $(\theta_t)_{t \in \mathbb{N}}$ be a predictable process, meaning θ_t is measurable with respect to $\mathcal{F}_{t-1}$, such that $\theta_t \geq \theta_0$ for all $t \in \mathbb{N}$ and define

$$\Lambda_t^{\text{plugin}} = \prod_{i=1}^{t} \frac{p_{\theta_i}(X_i)}{p_{\theta_0}(X_i)}. \quad (19)$$

The predictable plugin method replaces the numerator of the likelihood ratio the density of a fixed alternative by an estimated density p_{θ_t} that is based on the previous data $X_1, \ldots, X_{t-1}$. By replacing Λ_t with $\Lambda_t^{\text{plugin}}$ in (8), it follows immediately that $(\Lambda_t^{\text{plugin}})$ defines a test martingale for H_0. Hence, the application of Algorithm 1 is straightforward: at each $t \in \mathbb{N}$, we calculate a boosting factor using (16) by replacing θ_1 with θ_t.

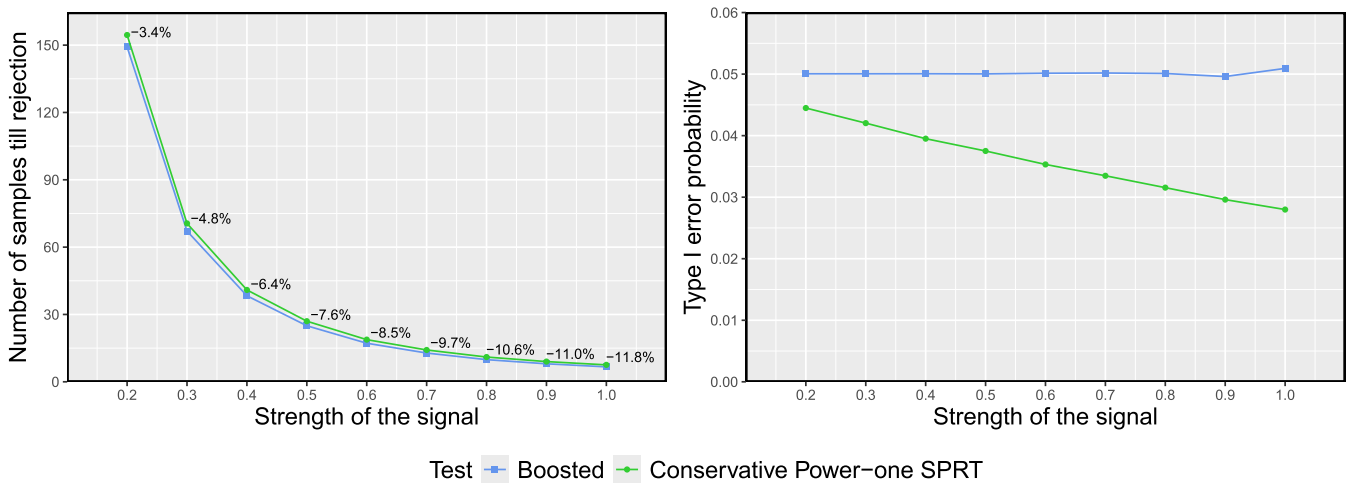


Fig. 1. Comparison of sample size and type I error between the SPRT and the boosted SPRT in a simple Gaussian testing setup for different signal strengths and $\alpha = 0.05$. The boosted SPRT saves $3.4 - 11.8\%$ of sample size compared to the SPRT and exhausts the significance level. The SPRT is becoming increasingly conservative for stronger signals.

A simple predictable plugin is given by the smoothed maximum likelihood estimator

$$\theta_t = \max\left(\frac{\theta_0 + \sum_{i=1}^{t-1} X_i}{t}, \theta_0\right). \tag{20}$$

In Figure 2 we repeated the experiment from Figure 1 in Section III-A but assumed that the alternative is not specified in advance using the predictable parameter in (20). Unsurprisingly, the sample size required by both methods is larger than when the alternative was prespecified. However, the gain in sample size due to boosting is even larger in the case of the composite alternative, saving 6.3% to 13.9% of the sample size compared to the usual SPRT.

C. Handling Composite Nulls in One-Sided Testing Problems

An even more common testing setup in practice is

$$H_0 : \theta^* \leq \theta_0 \quad \text{vs.} \quad H_1 : \theta^* > \theta_0 \quad (\theta_0 \in \Theta), \tag{21}$$

meaning we have a composite null hypothesis and a composite alternative. Throughout this section, we assume that for all $\theta', \theta'' \in \Theta$ with $\theta'' \geq \theta'$,

$$\lambda(x) = \frac{p_{\theta''}(x)}{p_{\theta'}(x)} \text{ is nondecreasing in } x. \tag{22}$$

Robbins and Siegmund [23] showed that under this assumption the process $(\Lambda_t^{\text{plugin}})$ defined in Equation (19) also provides a valid sequential test for the composite null hypothesis above. In the following, we show that the same holds for our boosted process.

Proposition 3: Consider the one-sided testing problem (21) under the monotone likelihood ratio property (22) and let $\lambda_t(X_t) = \frac{p_{\theta_t}(X_t)}{p_{\theta_0}(X_t)}$ for some predictable parameter $\theta_t \geq \theta_0$. Then,

$$\mathbb{E}_{\theta_0}[T_\alpha(b_t\lambda_t(X_t); M)|\Lambda_{t-1}^{\text{boost}}] \leq 1$$
$$\implies \mathbb{E}_{\theta}[T_\alpha(b_t\lambda_t(X_t); M)|\Lambda_{t-1}^{\text{boost}}] \leq 1$$

for any $\theta \leq \theta_0$, $\alpha \in (0, 1)$, $M > 0$ and $b_t \geq 1$.

To summarize, for the composite testing problem above and under the monotone likelihood ratio property, we can do exactly the same as described in Section III-B for the simple null hypothesis without inflating type I error.

Remark 4: An important class satisfying property (22) is the one-parameter exponential family, where each density is given by $p_\theta(x) = h(x)\exp(T(x)\eta(\theta) - A(\theta))$ with h, T, η and A being known functions. It is easy to see that (22) is fulfilled for the sufficient statistic $T(x)$, if $\eta(\theta)$ is nondecreasing in θ.

IV. Nontrivial Extensions and Applications

We now demonstrate some nontrivial extensions and applications of our boosting method. First, we show how it can be used for tighten confidence sequences in Section IV-A. Then, we exemplify its use in sampling without replacement situations (Section IV-B) and with conformal martingales (Section IV-C) for testing the assumption of i.i.d. data.

A. Confidence Sequences

A $(1-\alpha)$-level confidence sequence (CS) for $\theta^* \in \Theta$ is a sequence of sets $(C_t)_{t\in\mathbb{N}}$ such that

$$\mathbb{P}_{\theta^*}(\exists t \in \mathbb{N} : \theta^* \notin C_t) \leq \alpha. \tag{23}$$

Let τ_θ, $\theta \in \Theta$, be a sequential test for $H_0^\theta : \theta^* = \theta$ with property (6). Then

$$C_t = \{\theta \in \Theta : \tau_\theta > t\}$$

is a CS for θ^*. Let (M_t^θ) be the test supermartingale for H_0^θ that defines τ_θ, meaning $\tau_\theta = \inf\{t \in \mathbb{N} : M_t^\theta \geq 1/\alpha\}$. It follows immediately that $(C_t)_{t\in\mathbb{N}}$ can be boosted, meaning each C_t shrinks in size while maintaining (23), by boosting each of the martingales (M_t^θ), $\theta \in \Theta$, by Algorithm 1. However, the computational effort of comparing M_t^θ with $1/\alpha$ for every $\theta \in \Theta$ is extremely high or even infeasible. Due to

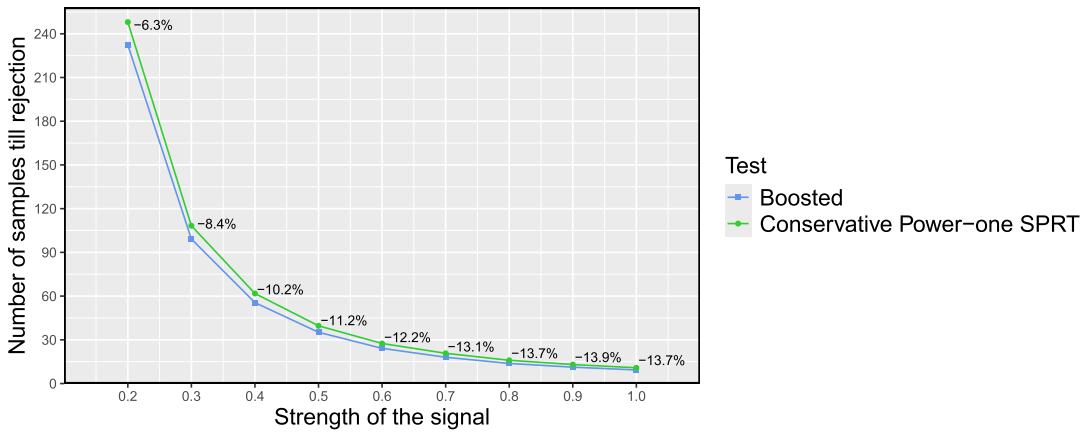


Fig. 2. Comparison of sample size between the SPRT with a predictable plugin and the boosted SPRT in a Gaussian testing setup with composite alternative for different signal strengths. The boosted SPRT saves $6.3 - 13.9\%$ of sample size compared to the SPRT.

computational convenience and interpretability, in practice one usually focuses on CSs that form an interval $C_t = (l_t, u_t)$. Often, two-sided CSs can then be obtained as the intersection of an upper and a lower $(1-\alpha/2)$-level CS. Hence, for now, let us for look a sequence $(l_t)_{t\in\mathbb{N}}$ such that

$$\mathbb{P}_{\theta^*}(\exists t \in \mathbb{N} : \theta^* < l_t) \leq \alpha.$$

In order to ensure that the resulting CS has this form, it is required that the martingales $(M_t^\theta)_{\theta\in\Theta}$ are nonincreasing in θ for every $t \in \mathbb{N}$. Now note that each boosting factor b_t^θ, $\theta \in \Theta$ and $t \in \mathbb{N}$, defined by Algorithm 1, is a nondecreasing function of the (boosted) supermartingale at the previous step $M_{t-1}^{\theta,\text{boost}}$. Therefore, the boosting factors $(b_t^\theta)_{\theta\in\Theta}$ are also nonincreasing in θ, provided that the distribution of (M_t^θ) under H_0^θ is the same for all $\theta \in \Theta$. This implies that also the boosted processes $(M_t^{\theta,\text{boost}})_{\theta\in\Theta}$ are nonincreasing in θ for each t such that the resulting CS provides indeed a lower bound. With this, the boosted CS can be easily determined computationally by setting l_t at each step $t \in \mathbb{N}$ to be the smallest $\theta \in \Theta$ such that $M_t^{\theta,\text{boost}} < 1/\alpha$. In Supplementary Material S.1, we illustrate this by boosting a CS for Gaussian means.

B. Sampling Without Replacement

Suppose we have some finite population $\{x_1, \ldots, x_N\}$ and draw samples

$$X_t | \{X_1, \ldots, X_{t-1}\} \sim \text{Uniform}(\{x_1, \ldots, x_N\} \setminus \{X_1, \ldots, X_{t-1}\})$$

sequentially and without replacement with the goal of testing the hypothesis

$$H_0 : \mu^* \leq \mu_0 \quad \text{vs.} \quad H_1 : \mu^* > \mu_0,$$

where $\mu^* = \frac{1}{N}\sum_{i=1}^N x_i$, because the effort of drawing all samples is too high. In this section, we focus on binary variables X_i. For example, this scenario is encountered in Risk-limiting audits (RLA) used for auditing the results of an election [25] or when checking whether the permutation p-value is below the predefined level α [26]. Also note that this testing scenario is very different from the classical i.i.d. Bernoulli SPRT setting, since the conditional mean $C_i(\mu_0) = \frac{N\mu_0 - \sum_{j=1}^{i-1} X_j}{N-i+1}$ of $X_i | X_1, \ldots, X_{i-1}$, if the mean of $\{x_1, \ldots, x_N\}$ is μ_0, changes after each sample. In the following, we show how the state-of-the-art method for this setting can be improved by our boosting technique.

Waudby-Smith et al. [25] proposed the RiLACS test martingale

$$M_t = \prod_{i=1}^t (1 + \lambda_i(X_i - C_i(\mu_0))), \tag{24}$$

where $\lambda_i \in \left[0, \frac{1}{C_i(\mu_0)}\right]$ is a hyperparameter measurable with respect to $\mathcal{F}_{i-1}$. Hence, each individual factor of (M_t) is just given by $L_t = 1 + \lambda_i(X_i - C_i(\mu_0))$. Under the null hypothesis and conditional on $\mathcal{F}_{t-1}$, we have that $L_t = 1 + \lambda_i(1 - C_i(\mu_0))$ with probability at most $C_i(\mu_0)$ and $L_t = 1 - \lambda_i C_i(\mu_0)$ with probability at least $1 - C_i(\mu_0)$. This implies that we can determine exact boosting factors b_t by

$$b_t = \frac{1 - C_i(\mu_0)/(M_{t-1}^{\text{boost}}\alpha)}{(1 - C_i(\mu_0))(1 - \lambda_i C_i(\mu_0))},$$

if $M_{t-1}^{\text{boost}}(1 + \lambda_i(1 - C_i(\mu_0))) \geq 1/\alpha$, and $b_t = 1$, otherwise.

Waudby-Smith et al. [25] proposed to choose $\lambda_i = \min(1/C_i(\mu_0), 2(2\mu_1 - 1))$, if the alternative $H_1 : \mu^* = \mu_1$ is known in advance. For example, this is the case in Risk-limiting audits. In Figure 3, we compare the required sample size of RiLACS and its boosted improvement for $\mu_0 = 0.5$ and $\mu_1 = 0.55$ for different total number of samples N at level $\alpha = 0.01$. The gain in sample size obtained by boosting is small but visible. This is not surprising since the RiLACS test martingale (24) only overshoots slightly at $1/\alpha$ due to the binary data. However, since b_t has a closed form, this improvement is essentially free.

C. Conformal martingales

A common assumption in statistics has been that the data is generated independently from the same distribution, also called i.i.d. (independent and identically distributed). Here,

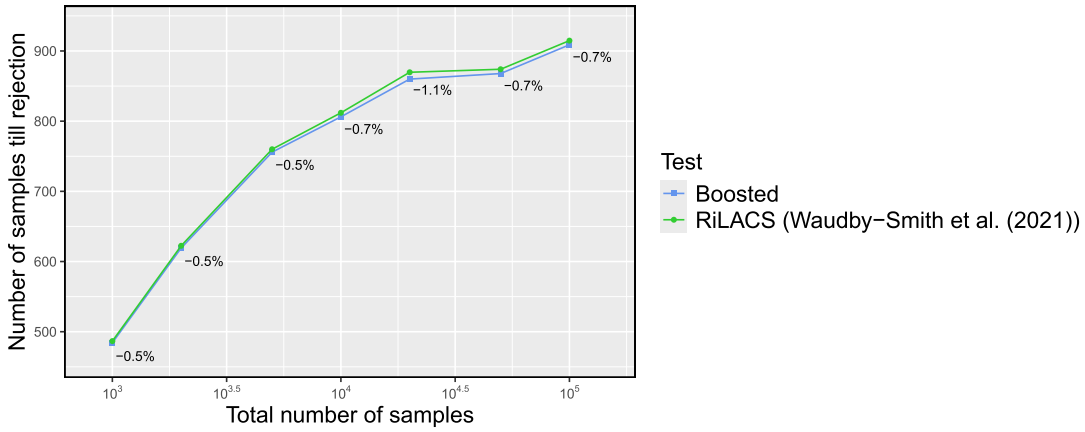


Fig. 3. Comparison of the required sample size by RILACS [25] and its boosted improvement in a sampling without replacement setting for different total number of samples. Boosting leads to a small but consistent gain in sample size.

we consider testing this assumption sequentially, meaning to test the composite null hypothesis

$$H_0 : X_1, X_2, \dots \text{ are i.i.d.}$$

From a single data sequence it is impossible to distinguish whether the data is i.i.d. or exchangeable [27], where $(X_t)_{t\in\mathbb{N}}$ is called exchangeable if $(X_1, \dots, X_t)$ has the same joint distribution as $(X_{\sigma(1)}, \dots, X_{\sigma(t)})$ for every $t \in \mathbb{N}$ and permutation σ of the first t indices. Hence, H_0 can be formulated equivalently as

$$H_0 : X_1, X_2, \dots \text{ are exchangeable.}$$

Vovk's conformal martingale approach [28] allows to reduce H_0 to a simple null hypothesis. This makes it not only possible to test H_0 sequentially using a test martingale, but also to apply our boosting technique from Algorithm 1. We first recap the method by Vovk and then demonstrate how we can apply our boosting approach on top of it.

Let A be a *nonconformity measure*, which maps observations $(x_1, \dots, x_t)$ of any length to a sequence of numbers $(z_1, \dots, z_t) \in \mathbb{R}^t$ such that for any t and any permutation σ,

$$\begin{aligned} &A(x_1, \dots, x_t) = (z_1, \dots, z_t) \\ \implies &A(x_{\sigma(1)}, \dots, x_{\sigma(t)}) = (z_{\sigma(1)}, \dots, z_{\sigma(t)}). \end{aligned}$$

With this, for each $t \in \mathbb{N}$, one can calculate a "conformal p-value" by

$$p_t = \frac{|\{i \le t : z_i > z_t\}| + \theta_t |\{i \le t : z_i = z_t\}|}{t},$$

where $(z_1, \dots, z_t) = A(x_1, \dots, x_t)$ and θ_t is randomly and independently sampled from $U[0,1]$. Vovk showed that the p-values $p_1, p_2, \dots$ are i.i.d. from $U[0,1]$ under the null hypothesis H_0, and proposed to collect evidence against H_0 by forming a *conformal martingale*

$$M_t = \prod_{i=1}^{t} f_i(p_i),$$

where the 'betting function' $f_i : [0,1] \to [0,\infty]$, $i \in \mathbb{N}$ may depend on the previous p-values and satisfies $\int_0^1 f_i(u)du = 1$. It is easy to check that this ensures $\mathbb{E}_0[f_t(p_t)|\mathcal{F}_{t-1}] = 1$, where $\mathcal{F}_{t-1} = \sigma(p_1, \dots, p_{t-1})$, such that (M_t) is indeed a test martingale for H_0 and $f_i(p_i)$ are the corresponding individual factors. Note that (M_t) may not be a martingale with respect to the filtration generated by the observations $X_1, X_2, \dots$, but it surely is with respect to the filtration generated by the p-values $p_1, p_2, \dots$. Since we know that the p-values $p_1, p_2, \dots$ are i.i.d. uniformly distributed on $U[0,1]$ under H_0, we can easily determine the distribution of $f_i(p_i)$ conditional on the previous p-values such that Algorithm 1 can be applied.

For example, a concrete betting function proposed by Vovk is $f^\kappa(u) = \kappa u^{\kappa-1}, \quad \kappa \in (0,1)$. At each step $t \in \mathbb{N}$, setting $\mathbb{E}_0[T_\alpha(b_t f_t(p_t); M_{t-1}^{\text{boost}})|\mathcal{F}_{t-1}] = 1$ yields the equation

$$\begin{aligned} &b_t \max\left(1 - \left(\frac{1/(\alpha M_{t-1}^{\text{boost}})}{b_t \kappa}\right)^{\frac{\kappa}{\kappa-1}}, 0\right) \\ &+ \frac{1}{\alpha M_{t-1}^{\text{boost}}} \min\left(\left(\frac{1/(\alpha M_{t-1}^{\text{boost}})}{b_t \kappa}\right)^{\frac{\kappa}{\kappa-1}}, 1\right) = 1, \end{aligned}$$

which can be solved for b_t. For example, $\alpha = 0.05$, $M_{t-1}^{\text{boost}} = 1$ and $\kappa = 0.5$ lead to a boosting factor of $b_t = 1.013$. The boosting factor is decreasing in κ such that we obtain a boosting factor of $b_t = 2.199$ for $\kappa = 0.1$. In practice, it can be useful to choose a predictable sequence of parameters $(\kappa_t)_{t\in\mathbb{N}}$ and employ the betting function f^{k_t} for the p-value p_t [28]. However, this poses no problem for our boosting algorithm (see Section III-B).

V. Challenging Wald's two-sided Approximate SPRT

While the previous sections focused on power-one SPRTs ($\beta = 0$) where we only stop to reject the null but not to accept it, we now consider the case of $\beta > 0$, where we also wish to stop if it is unlikely that we would reject the null if we continued testing. We will show how we can use a boosting technique similar to that in Section II to construct powerful sequential tests that allow to "stop for futility" (stop and accept the null).

Our main comparison point will be Wald's approximate two-sided SPRT, which uses thresholds of $(1-\beta)/\alpha$ and

$\beta/(1-\alpha)$. Importantly, out of the three desiderata of (a) type I error control at α, (b) type II error control at β, (c) optimal sample size, these approximate thresholds guarantee none of the three properties. In this section, we will first develop a general method with valid type I error control for test supermartingales, and then show how the stop for futility can be chosen to obtain tight type I and type II error control in the SPRT case. The latter method often outperforms Wald's approximate SPRT in the sense that while guaranteeing (a) and (b), our method also manages to typically/often stop sooner than Wald's (though not guaranteeing (c)).

A. Two-Sided Stopping With Type I Error Control for supermartingales

Suppose our test is based on a test (super)martingale (M_t), such as the likelihood ratio. We consider stopping for futility if $M_t \leq \nu_t$, where $\nu_t \leq 1/\alpha$ is a predictable parameter, meaning ν_t is measurable with respect to $\mathcal{F}_{t-1}$. For example, Wald's approximation suggests to set $\nu_t = \frac{\beta}{1-\alpha}$. Note that stopping for futility can never increase the type I error, but it may hurt the power. Below, we will show how we can gain back part of that power.

If $M_t > 0$, we have $\mathbb{E}_0[M_{t'}|\mathcal{F}_t] = M_t > 0$ for all $t' > t$. Hence, if we stop when our test martingale is strictly positive, we make a conservative decision since there is a positive probability of rejecting if we continue sampling. Therefore, our approach is to form a truncated test (super)martingale (M_t^{trunc}) such that $M_t \leq \nu_t$ implies that $M_t^{\text{trunc}} = 0$. To achieve this, we extend our truncation function $T_\alpha(x; M)$ from (10) as follows:

$$T_\alpha(x; M, \nu) = \begin{cases} 0, & \text{if } Mx \leq \nu \\ x, & \text{if } \nu < Mx \leq \frac{1}{\alpha} \\ \frac{1}{M\alpha}, & \text{if } Mx > \frac{1}{\alpha}. \end{cases} \tag{25}$$

We can use $T_\alpha(x; M, \nu)$ for boosting in the same manner as we did with $T_\alpha(x; M)$ for the one-sided stopping boundary by choosing a boosting factor $b_t \geq 1$ such that

$$\mathbb{E}_0[T_\alpha(b_t L_t; M_{t-1}^{\text{boost}}, \nu_t)|\mathcal{F}_{t-1}] \leq 1, \tag{26}$$

where $M_{t-1}^{\text{boost}} = \prod_{i=1}^{t-1} T_\alpha(b_i L_i; M_{i-1}^{\text{boost}}, \nu_i)$. Note that $T_\alpha(x; M, \nu) = T_\alpha(x; M)$, if $\nu = 0$. Furthermore, $T_\alpha(x; M, \nu) \leq T_\alpha(x; M)$ for all $x \in \mathbb{R}_{\geq 0}$, implying that the boosting factors obtained by $T_\alpha(x; M, \nu)$ are larger than the boosting factors computed with $T_\alpha(x; M)$. The following theorem extends Theorem 1 by including a specific stop for futility.

Theorem 4: Let $\nu_t^M \leq 1/\alpha$, $t \in \mathbb{N}$, be a predictable parameter and $\nu_t = \min\left(\nu_t^M \prod_{i=1}^t b_i, 1/\alpha\right)$. Furthermore, let (M_t) be any test (super)martingale and (M_t^{boost}) be the process obtained by applying Algorithm 1 with $T_\alpha(x; M_{t-1}^{\text{boost}}, \nu_t)$ and define the sequential tests $\delta_M = \mathbb{1}\{M_{\tau_M} \geq 1/\alpha\}$ and $\delta_{\text{boost}} = \mathbb{1}\{M_{\tau_{\text{boost}}}^{\text{boost}} \geq 1/\alpha\}$, where $\tau_M = \inf\{t \in \mathbb{N} : M_t \geq 1/\alpha \vee M_t \leq \nu_t^M\}$ and $\tau_{\text{boost}} = \inf\{t \in \mathbb{N} : M_t^{\text{boost}} \geq 1/\alpha \vee M_t^{\text{boost}} \leq \nu_t\}$. Then $M_t^{\text{boost}} \geq M_t$ for all $t < \tau_{\text{boost}}$, $\delta_{\text{boost}} \geq \delta$ and $\tau_{\text{boost}} \leq \tau_M$. Furthermore, (M_t^{boost}) is a test supermartingale. Therefore, $\mathbb{P}_0(\delta_{\text{boost}} = 1) \leq \alpha$ and $\mathbb{E}_0[M_\tau^{\text{boost}}] \leq 1$ for all stopping times τ.

Note that in the above theorem ν_t is a function of b_t. This ensures that our boosted martingale (M_t^{boost}) always stops in cases where the initial martingale (M_t) stops for futility, which allows to prove that $\tau_{\text{boost}} \leq \tau_M$. However, we think in practice both scenarios are reasonable. We could either want to provide a stop for futility ν_t directly (independently of b_t) for M_t^{boost} or provide a stop for futility ν_t^M for M_t (independently of b_t) and then set $\nu_t = \min\left(\nu_t^M \prod_{i=1}^t b_i, 1/\alpha\right)$. For the rest of this subsection, we treat ν_t as independent of b_t. However, all results shown apply equivalently for $\nu_t = \min\left(\nu_t^M \prod_{i=1}^t b_i, 1/\alpha\right)$, where ν_t^M does not dependent on b_t.

It turns out that in the two-sided case Proposition 2 no longer holds, meaning we cannot always find a b_t which solves (26) with an equality.

Example 1: Suppose we have i.i.d. binary data $X_1, X_2, \ldots$ and $\mathbb{E}_0[X_t] = 1/3$ and $\mathbb{E}_1[X_t] = 2/3$. Then the likelihood ratio $\lambda(X_t) = 2$, if $X_t = 1$, and $\lambda(X_t) = 1/2$ otherwise. If $\alpha = 0.05$, $\Lambda_{t-1}^{\text{boost}} = 8$ and $\nu_t = 7$. Then $\mathbb{E}_0[T_\alpha(b_t L_t; \Lambda_{t-1}^{\text{boost}}, \nu_t)|\mathcal{F}_{t-1}] \leq 1/3 \cdot 20/8 < 1$ for all $1 \leq b_t \leq 7/4$ and $\mathbb{E}_0[T_\alpha(b_t L_t; \Lambda_{t-1}^{\text{boost}}, \nu_t)|\mathcal{F}_{t-1}] \geq 2/3 \cdot 7/8 + 1/3 \cdot 20/8 > 1$ for all $b_t > 7/4$.

The problem in Example 1 is that $T_\alpha(x; M, \nu)$ is not continuous in x and that the data is binary, creating a jump in the expectation of the boosted (super)martingale factor. In Supplementary Material S.3, we discuss how external randomization could be used to avoid conservatism with discrete data when using a stop for futility and in the following proposition we show that Proposition 2 can at least be extended to two-sided stopping for continuously distributed martingale factors.

Proposition 5: If $L_t = M_t/M_{t-1}$ is continuously distributed under $\mathbb{P}_0$ and $M_{t-1}^{\text{boost}} \in (0, \alpha^{-1})$, then there exists a $b_t^* \geq 1$ such that

$$\mathbb{E}_0[T_\alpha(b_t^* L_t; M_{t-1}^{\text{boost}}, \nu_t)|\mathcal{F}_{t-1}] = 1. \tag{27}$$

Further, for every other $b_t \geq 1$ that satisfies (26), we have that $T_\alpha(b_t^* L_t; M_{t-1}^{\text{boost}}, \nu_t) \geq T_\alpha(b_t L_t; M_{t-1}^{\text{boost}}, \nu_t)$ $\mathbb{P}_0$-almost surely.

In the same manner as in (16), we can calculate for SPRTs:

$$\begin{aligned} &\mathbb{E}_{\theta_0}[T_\alpha(b_t \lambda(X_t); \Lambda_{t-1}^{\text{boost}}, \nu_t)|\Lambda_{t-1}^{\text{boost}}] \\ &= b_t \mathbb{P}_{\theta_1}\left(\frac{\nu_t}{b_t \Lambda_{t-1}^{\text{boost}}} < \lambda(X_t) \leq \frac{1}{b_t \alpha \Lambda_{t-1}^{\text{boost}}} \middle| \Lambda_{t-1}^{\text{boost}}\right) \\ &+ \frac{1}{\Lambda_{t-1}^{\text{boost}} \alpha} \mathbb{P}_{\theta_0}\left(\lambda(X_t) > \frac{1}{b_t \alpha \Lambda_{t-1}^{\text{boost}}} \middle| \Lambda_{t-1}^{\text{boost}}\right). \end{aligned}$$

The boosting factors obtained in the same setup as for Table II, but with a stop for futility at $\nu_t = 0.4$, is shown in Table III. Obviously, all boosting factors are larger when the stop for futility is included. Furthermore, especially in cases where the current martingale value $\Lambda_{t-1}^{\text{boost}}$ is small, including the stop for futility increases the boosting factor.

Remark 5: Note that our boosted SPRT could also be written as

$$\begin{aligned} &\text{reject } H_0, \text{ if } \Lambda_t \geq \gamma_1^t, \\ &\text{accept } H_0, \text{ if } \Lambda_t \leq \gamma_0^t, \\ &\text{continue sampling, if } \gamma_0^t < \Lambda_t < \gamma_1^t, \end{aligned}$$

TABLE III
BOOSTING FACTORS OBTAINED FOR $\alpha = 0.05$ AND $\nu_t = 0.4$, $t \in \mathbb{N}$, IN A SIMPLE GAUSSIAN TESTING PROBLEM WITH MEAN DIFFERENCE δ AND VARIANCE 1

δ \ $\Lambda_{t-1}^{\text{boost}}$	0.5	1	2	4	10
0.1	1.00895	1	1	1	1.00001
0.5	1.17964	1.01743	1.00026	1.00019	1.03019
1	1.21801	1.07547	1.02817	1.05651	1.37357
2	1.32013	1.38991	1.62094	2.21769	5.76214
3	2.73073	3.75762	6.00201	12.1467	68.8295

where $\gamma_1^t = \frac{1}{\alpha \prod_{i=1}^t b_i}$ and $\gamma_0^t = \frac{\nu_t}{\prod_{i=1}^t b_i}$ are predictable parameters. Such generalized SPRTs with non-constant boundaries over time were already introduced by [29], however, Weiss assumed that γ_0^t and γ_1^t, $t \in \mathbb{N}$, are prespecified constants.

Remark 6: Typically, the parameter ν_t is set to some value smaller than 1, as $\Lambda_t < 1$ implies that the current data favors the null hypothesis over the alternative. However, ν_t is a predictable parameter and one could also think of choosing it based on the current value of $\Lambda_{t-1}^{\text{boost}}$. For example, it might be reasonable to set $\nu_t = 8$ if $\Lambda_{t-1}^{\text{boost}} = 10$ to stop in case of decreasing evidence. In general, it is even possible to set $\nu_t > \Lambda_{t-1}^{\text{boost}}$.

B. Two-Sided Stopping With Type I, II Error Control for SPRTs

We know that boosting always ensures type I error control at the level α by Ville's inequality (7). Now we want to use the same approach to choose the stop for futility ν_t in a way that ensures valid type II error control at a level $\beta \in (0,1)$. The idea is to use a second SPRT for testing the alternative $H_1 : X_i \sim \mathbb{P}_1$ against the null hypothesis $H_0 : X_i \sim \mathbb{P}_0$ and stop for futility with our original test if this "inverse SPRT" rejects H_1. Hence, instead of using the likelihood ratio process (Λ_t) of $\mathbb{P}_1$ against $\mathbb{P}_0$, we use its inverse $\Lambda_t^{\text{inv}} = 1/\Lambda_t$. By Ville's inequality (7), we control the type II error for the original test of H_0 against H_1 at level β, if we stop for futility when the inverse likelihood ratio Λ_t^{inv} crosses $1/\beta$. However, as we demonstrated in the previous sections, Ville's inequality is loose if there is overshoot. Hence, to get tighter type II error control, we need to boost the inverse SPRT in the same manner as we did for the usual SPRT and include the stop for futility yielded by the inverse SPRT in the truncation function (25). Precisely, denoting $\lambda_t(X_t) = p_1(X_t)/p_0(X_t)$ at each step $t \in \mathbb{N}$, we choose $b_t \geq 1$ and $b_t^{\text{inv}} \geq 1$ as large as possible such that

$$\mathbb{E}_0\left[T_\alpha(b_t\lambda_t(X_t); \Lambda_{t-1}^{\text{boost}}, \nu_t)\middle|\mathcal{F}_{t-1}\right] \leq 1, \tag{28}$$

$$\mathbb{E}_1\left[T_\beta(b_t^{\text{inv}}/\lambda_t(X_t); \Lambda_{t-1}^{\text{boost,inv}}, \nu_t^{\text{inv}})\middle|\mathcal{F}_{t-1}\right] \leq 1, \tag{29}$$

where

$$\nu_t = \min(\beta \prod_{i=1}^t b_i \prod_{i=1}^t b_i^{\text{inv}}, 1/\alpha),$$

$$\nu_t^{\text{inv}} = \min(\alpha \prod_{i=1}^t b_i \prod_{i=1}^t b_i^{\text{inv}}, 1/\beta),$$

$$\Lambda_t^{\text{boost}} = \prod_{i=1}^t T_\alpha(b_i\lambda_i(X_i); \Lambda_{i-1}^{\text{boost}}, \nu_i),$$

$$\Lambda_t^{\text{boost,inv}} = \prod_{i=1}^t T_\alpha(b_i^{\text{boost}}/\lambda_i(X_i); \Lambda_{i-1}^{\text{boost,inv}}, \nu_i^{\text{inv}}).$$

The definition of ν_t ensures that the original SPRT only stops for futility when the boosted inverse SPRT stops for rejection. We next use this to yield valid error control.

Theorem 6: Let the boosting factors $b_1, b_2, \ldots$ and inverse boosting factors $b_1^{\text{inv}}, b_2^{\text{inv}}, \ldots$ be chosen such that (28) and (29) are fulfilled for each $t \in \mathbb{N}$. Then the boosted SPRT given by $\delta^{\text{boost}} = \mathbb{1}\{\Lambda_{\tau_{\text{boost}}}^{\text{boost}} \geq 1/\alpha\}$ with $\tau_{\text{boost}} = \inf\{t \in \mathbb{N} : \Lambda_t^{\text{boost}} \geq 1/\alpha \vee \Lambda_t^{\text{boost}} \leq \nu_t\}$ controls the type I error at level α and the type II error at level β.

Similar to before, it follows that the system above can often be solved with equalities.

Proposition 7: If $\lambda_t(X_t)$ is continuously distributed under $\mathbb{P}_0$ and $\Lambda_{t-1}^{\text{boost}} \in (0, \alpha^{-1}), \Lambda_{t-1}^{\text{boost,inv}} \in (0, \beta^{-1})$, then there exist b_t^* and $b_t^{*,\text{inv}}$ satisfying (28) and (29) with equality.

Note that even if (28) and (29) are satisfied with an equality, the error control can still be slightly conservative, since it may happen that

$$\beta \prod_{i=1}^t b_i \prod_{i=1}^t b_i^{\text{inv}} > 1/\alpha,$$

$$\Leftrightarrow \alpha \prod_{i=1}^t b_i \prod_{i=1}^t b_i^{\text{inv}} > 1/\beta,$$

which means that the rejection region of the boosted SPRT and the boosted inverse SPRT overlap. This is why we set $\nu_t = 1/\alpha$ and $\nu_t^{\text{inv}} = 1/\beta$ in this case which ensures valid type I and type II error control but avoids the simultaneous acceptance and rejection of both hypotheses (in general, we can choose smaller ν_t and ν_t^{inv} than given above without violating error control). However, this conservatism is expected to be small, particularly if α and β are small. Next, we show that the boosted two-sided SPRT always beats Wald's conservative SPRT in terms of sample size.

Proposition 8: Let the boosting factors $b_1, b_2, \ldots$ and inverse boosting factors $b_1^{\text{inv}}, b_2^{\text{inv}}, \ldots$ be chosen such that (28) and (29) are fulfilled for each $t \in \mathbb{N}$. Furthermore, let $\tau_{\text{boost}} = \inf\{t \in \mathbb{N} : \Lambda_t^{\text{boost}} \geq 1/\alpha \vee \Lambda_t^{\text{boost}} \leq \nu_t\}$ be stopping time of the boosted SPRT and $\tau_{\text{cons}} = \inf\{t \in \mathbb{N} : \Lambda_t \geq 1/\alpha \vee \Lambda_t \leq \beta\}$ of Wald's conservative SPRT. Then $\tau_{\text{boost}} \leq \tau_{\text{cons}}$.

Note that it is not possible to generally improve Wald's approximate SPRT, since it does not provide valid type I and type II error control. However, in Figure 4 we compare the sample size, the type II and the type I error probability of the SPRT with Wald's approximated and conservative thresholds to our boosted method for different parameters β in the same Gaussian simulation setup as considered in Section III-A. The desired type I error probability was set to $\alpha = 0.05$ and the strength of the signal to $\delta = 0.3$. The boosted method reduced the required sample size in all cases. Furthermore, the boosted SPRT nearly exhausts the type I error and type II error probability, while even Wald's approximate SPRT is conservative in this case. For all β, one boosted SPRT, includ-

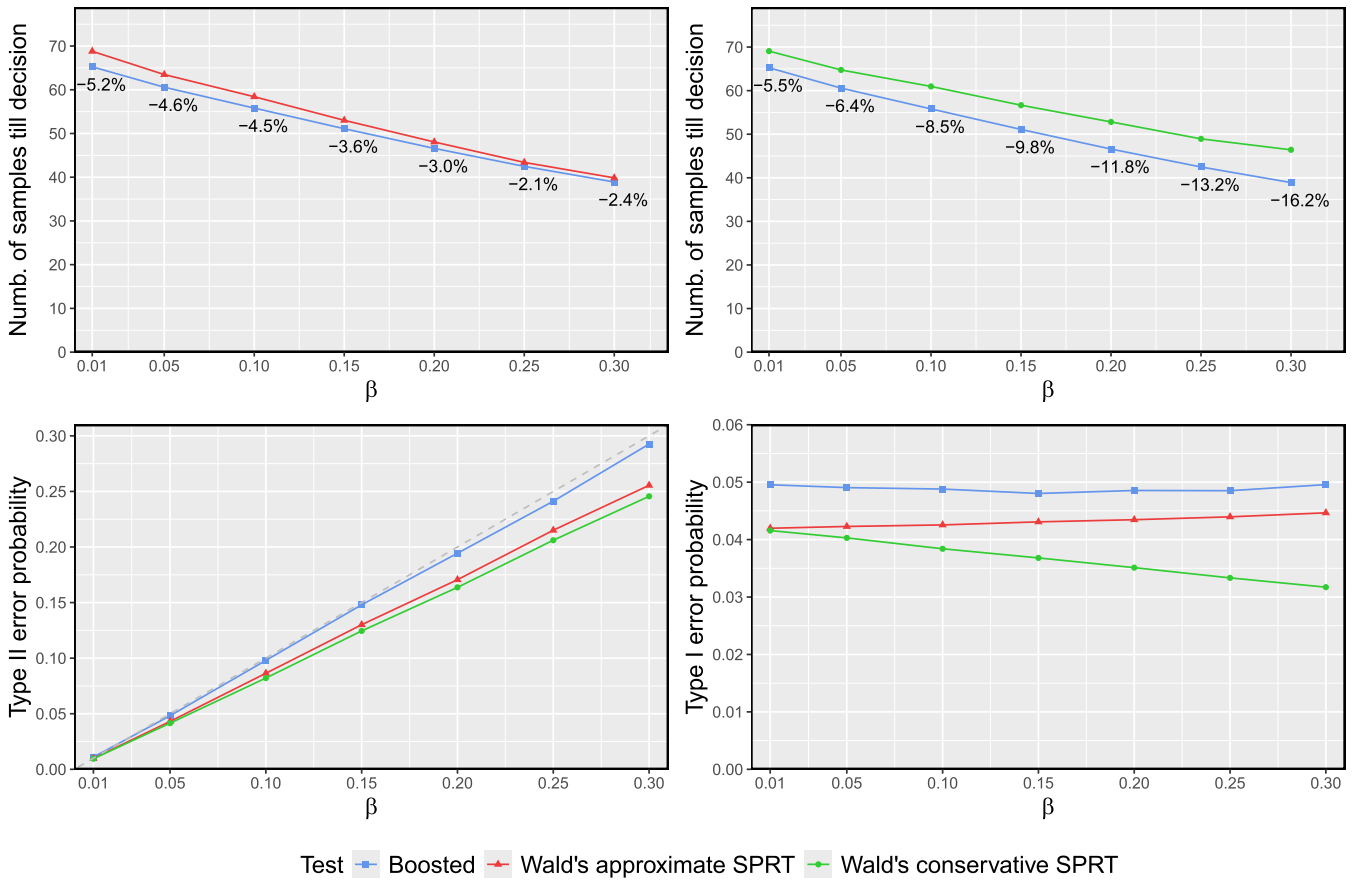


Fig. 4. Comparison of sample size, type II and type I error probability between Wald's approximate SPRT, Wald's conservative SPRT and the boosted SPRT in a simple Gaussian testing setup for different desired type II error probabilities (β) and $\alpha = 0.05$. Wald's conservative SPRT and our boosted method provably control the type I and type II error, while the SPRT with Wald's approximated thresholds does not. Furthermore, the boosted SPRT saves sample size and nearly exhausts the error probabilities.

ing the numerical calculation of the boosting factors, needed less than 0.3 seconds to compile on average. In Supplementary Material S.2 we repeat this experiment for $\alpha = 0.01$ instead of $\alpha = 0.05$, which leads to similar results. In Supplementary Material S.4, we give more details on our implementation of this boosted two-sided SPRT.

Remark 7: In order to avoid the simultaneous solving for b_t and b_t^{inv}, one could also just set $\nu_t' = \min\left(\beta \prod_{i=1}^t b_i, 1/\alpha\right)$ and choose $b_t \geq 1$ as large as possible such that

$$\mathbb{E}_0\left[T_\alpha(b_t \lambda_t(X_t); \Lambda_{t-1}^{\text{boost}}, \nu_t') \middle| \mathcal{F}_{t-1}\right] \leq 1.$$

Since $\nu_t' \leq \nu_t$ for ν_t in (28), this would lead to valid type I and type II error control. However, the type II error control would usually be more conservative than with the proposed approach.

C. Type I,II Error Control for One-Sided Composite Hypotheses

Consider a one-sided testing problem similar to that in Section III-C, but with a separation between the null hypothesis and alternative

$$H_0 : \theta^* \leq \theta_0 \quad \text{vs.} \quad H_1 : \theta^* \geq \theta_1 \quad (\theta_0 < \theta_1). \tag{30}$$

If the monotone likelihood ratio property (22) is satisfied, we can ensure type I and type II error control by using the boosted SPRT of θ_1 vs. θ_0.

Proposition 9: Consider the one-sided testing problem with separation (30) under the monotone likelihood ratio property (22). Then the boosted SPRT δ^{boost} described in Theorem 6 with $\mathbb{P}_1 = \mathbb{P}_{\theta_1}$ and $\mathbb{P}_0 = \mathbb{P}_{\theta_0}$ provides type I error control for all $\theta \leq \theta_0$ and type II error control for all $\theta \geq \theta_1$.

VI. Discussion

A. Testing a forecaster

The classical SPRT by Wald [1], [2] assumes that the null hypothesis and the alternative hypothesis are specified in advance in the form of simple distributions $\mathbb{P}_0$ and $\mathbb{P}_1$, respectively. However, in Section III-B we have seen that the SPRT remains even valid if we replace the fixed alternative $\mathbb{P}_1$ with a predictable plugin $\mathbb{P}_1^t$ that depends on the data $X_1, \ldots, X_{t-1} \in \mathcal{X}$ observed so far. In general, the null distribution $\mathbb{P}_0^t$ and the alternative distribution $\mathbb{P}_1^t$ considered at each step $t \in \mathbb{N}$ can be chosen freely based on the previous data $X_1, \ldots, X_{t-1}$ and some additional information $Z_1, \ldots, Z_t \in$

$\mathcal{Z}$. Shafer [30] formulated this as a forecaster-skeptic testing protocol, where the forecaster announces a probability distribution $\mathbb{P}_0^t$ at each step t and a skeptic tests the forecaster by specifying an alternative $\mathbb{P}_1^t$. For example, one can think of the forecaster as a weather forecaster, which is tried to be invalidated by a skeptic. We summarized this general testing protocol in Algorithm S.1 (see supplement).

The extension of our boosting technique (Algorithm 1) to the general testing problem in Algorithm S.1 is straightforward, since at each step t we only require the current martingale value $M_{t-1} = \Lambda_{t-1}$ and the distribution of the next factor $L_t = \frac{p_1^t(X_t)}{p_0^t(X_t)}$ under the null hypothesis $\mathbb{P}_0^t$, where p_0^t and p_1^t are the densities of $\mathbb{P}_0^t$ and $\mathbb{P}_1^t$, respectively.

B. Other Approaches to Avoid Wald's Approximated thresholds

The problem of "overshoot" in the SPRT was already noted by Wald [1], [2] and examined in many previous works. One line of work approximates the exact type I and type II error of SPRTs based on Brownian motion and renewal theory [5], [31], [32], [33]. This approach was particularly driven by Siegmund, who gives a comprehensive overview in his book [5]. The provided approximations are asymptotically (for $\alpha \to 0$ and $\beta \to 0$) correct under certain assumptions including i.i.d. test statistics. Furthermore, using the Markov property of the likelihood ratio in simple i.i.d. settings, one can obtain the exact operating characteristics of the SPRT by solving Fredholm integral equations (see, e.g., [34], [35], [36]).

In Figures 1 and 4, we compare our boosted SPRT to the SPRT with Siegmund's approximations in a simple i.i.d. Gaussian testing setup $H_0 : X_t \sim \mathcal{N}(0,1)$ vs. $H_1 : X_t \sim \mathcal{N}(\mu_1, 1)$, where $\mu_1 > 0$ is the strength of the signal. Siegmund's approximations suggest to choose $\gamma_1 = \frac{1-\beta}{\alpha \exp(\mu_1 \rho)}$ and $\gamma_0 = \frac{\beta \exp(\mu_1 \rho)}{1-\alpha}$, where $\rho = 0.583$. In Figure 5, we compare the SPRT with these approximated thresholds to our boosted SPRT in the power-one ($\beta = 0$) case and in Figure 6 for different desired type II errors β. In both cases $\alpha = 0.05$ and the implementation of our boosted SPRT is exactly the same as in Figures 1 and 4. In this case we decided to use Siegmund's approximations for comparison, since they are easier to apply than the integral equation approach and the simulations showed that they are sufficiently accurate.

It is easy to see that Siegmund's SPRT outperformed our boosted SPRT in all considered cases. While the difference in the power-one scenario is hardly visible, the gain in sample size for larger β can be substantial. This is not surprising, since due to the result by [4] (see (3)), the SPRT with exact thresholds is optimal in terms of expected sample size. Therefore, if sufficiently accurate approximations of the exact thresholds are available, e.g., if Siegmund's theory applies or the Fredholm integral equations hold, we recommend the use of the SPRT with these thresholds. However, the advantage of our proposed boosting technique is its much broader applicability. For example, the aforementioned approaches cannot be applied to the one-sided SPRT without separation (Section III-B) or any of the applications provided in Section IV, since the employed test statistics are not independent (if $(\delta_t)_{t\in\mathbb{N}}$ and $(\kappa_t)_{t\in\mathbb{N}}$ are predictable in Sections S.1 and IV-C). In contrast, we do not make any separation or independence assumptions and our method provides non-asymptotic as well as non-approximate error control. and our method provides non-asymptotic as well as non-approximate error control.

To enable a more detailed and complete comparison of the considered methods, Table IV displays some of the results in comparison with Wald's SPRTs and also shows the expected sample size under the null distribution. To achieve higher accuracy, these results were obtained using 100 000 simulation runs.

Another approach is to truncate the SPRT by prespecifying a maximum number of observations N. For simple distributions of the likelihood ratio and sufficiently small N it is then possible to calculate the type I and type II error for given thresholds exactly [37], which can be used to find appropriate bounds for the SPRT. Such an approach is frequently employed in group sequential trials [38], [39]. For more complex distributions one could approximate the error probabilities using simulations. This often leads to considerable computational effort, which sometimes can be reduced using importance sampling [21].

One limitation all the above approaches have in common is that the test statistics and their distributions must be specified in advance. This is not possible in the general forecaster-skeptic testing protocol described in the previous section. Hence, while our boosting approach is applicable to such complex and adaptive testing protocols, all of the aforementioned methods fail. Furthermore, after calculating a boosting factor b_t with our method, one can always plug it back into the expectation ((12) in the one-sided or (28) and (29) in the two-sided case) and only accept the boosting factor if the expectation is indeed bounded by 1. If a boosting factor cannot be accepted, one can try to calculate another boosting factor or set $b_t = 1$ if no greater boosting factor can be found. In this way, one can always ensure that the resulting test is conservative, meaning that it provides the desired type I and type II error. This is neither possible with Siegmund's approach nor when using simulations.

Moreover, our approach provides a test supermartingale, which yields an e-value at stopping times [17], [30], [40]. For example, this can be important for optional continuation [30], [40], combining the evidence with other test results [41] and in multiple testing [42].

C. Promising Applications

Due to its tight error control in non-i.i.d. sequential settings, our boosted SPRT has the potential to become the state-of-the-art method for a variety of applications.

For example, in clinical trials we often encounter testing problems of the form $H_0 : \theta^* \leq 0$ vs. $H_1 : \theta^* > 0$, where θ^* describes the treatment effect of a new treatment compared to a control, and rigorous type I error control is required. Due to the composite alternative, we usually need to estimate θ^* based on the past data to apply the SPRT (see also Section III-B). Therefore, the likelihood ratio becomes a product of non-i.i.d. test statistics, which implies that all the alternative approaches (Siegmund, Fredholm) mentioned in the previous section do

TABLE IV

COMPARISON OF STOPPING TIMES AS WELL AS ACHIEVED TYPE I (α') AND TYPE II (β') ERROR PROBABILITIES BETWEEN WALD'S APPROXIMATE SPRT, WALD'S CONSERVATIVE SPRT, SIEGMUND'S SPRT AND THE BOOSTED SPRT IN A SIMPLE GAUSSIAN TESTING SETUP. THE INDEX VALUE GIVES THE MONTE CARLO STANDARD ERROR. SIEGMUND'S APPROXIMATE THRESHOLDS ARE THE BEST, BUT ARE ONLY COMPUTABLE IN VERY SPECIAL CASES, WHILE BOOSTED SPRT IS COMPETITIVE BUT MUCH MORE BROADLY APPLICABLE

Test	α	β	α'	β'	$\mathbb{E}_0[\tau]$	$\mathbb{E}_1[\tau]$
Wald's approx. SPRT	0.05	0.01	$0.0421_{2\cdot10^{-5}}$	$0.0081_{3\cdot10^{-4}}$	$98.06_{0.19}$	$69.34_{0.17}$
Wald's cons. SPRT	0.05	0.01	$0.0417_{2\cdot10^{-5}}$	$0.0078_{3\cdot10^{-4}}$	$99.21_{0.19}$	$69.62_{0.17}$
Siegmund's SPRT	0.05	0.01	$0.0500_{3\cdot10^{-5}}$	$0.0096_{3\cdot10^{-4}}$	$93.19_{0.18}$	$65.29_{0.16}$
Boosted SPRT	0.05	0.01	$0.0498_{2\cdot10^{-4}}$	$0.0094_{3\cdot10^{-4}}$	$94.54_{0.18}$	$65.91_{0.16}$
Wald's approx. SPRT	0.05	0.2	$0.0436_{7\cdot10^{-5}}$	$0.1689_{1\cdot10^{-3}}$	$34.33_{0.09}$	$48.16_{0.11}$
Wald's cons. SPRT	0.05	0.2	$0.0352_{5\cdot10^{-5}}$	$0.1620_{1\cdot10^{-3}}$	$36.11_{0.10}$	$52.86_{0.11}$
Siegmund's SPRT	0.05	0.2	$0.0501_{8\cdot10^{-5}}$	$0.1991_{1\cdot10^{-3}}$	$27.02_{0.09}$	$42.63_{0.10}$
Boosted SPRT	0.05	0.2	$0.0492_{3\cdot10^{-4}}$	$0.1914_{1\cdot10^{-3}}$	$30.16_{0.09}$	$46.60_{0.10}$

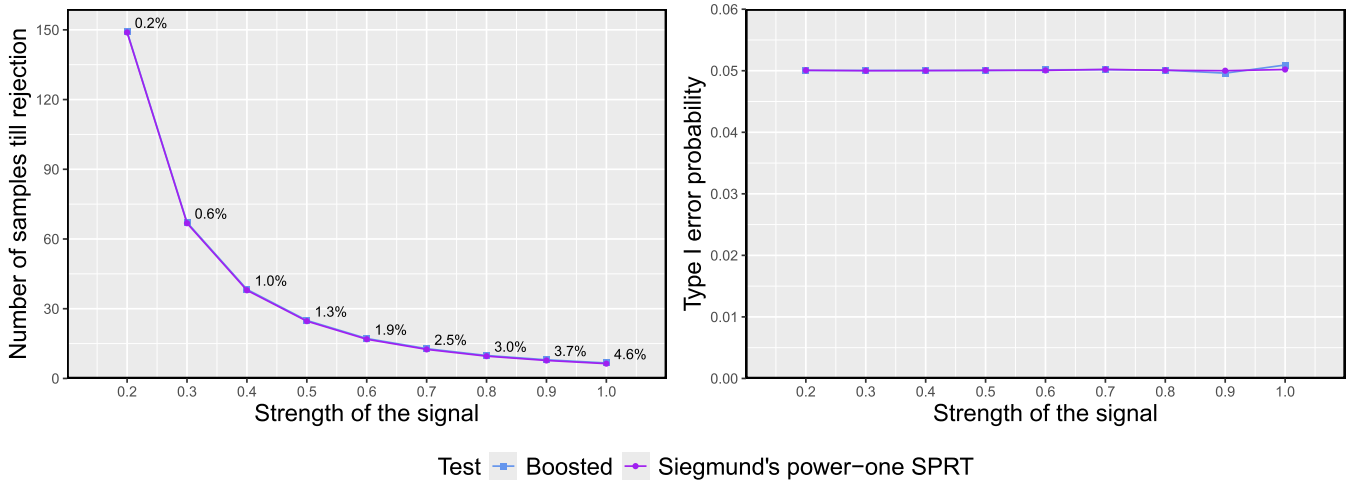


Fig. 5. Comparison of sample size and type I error between Siegmund's SPRT and the boosted SPRT in a simple Gaussian testing setup for different signal strengths and $\alpha = 0.05$. Siegmund's SPRT saves sample size compared to the boosted SPRT, but the gain is hardly visible. Siegmund's SPRT controls the type I error asymptotically, while the boosted SPRT provides non-asymptotic control.

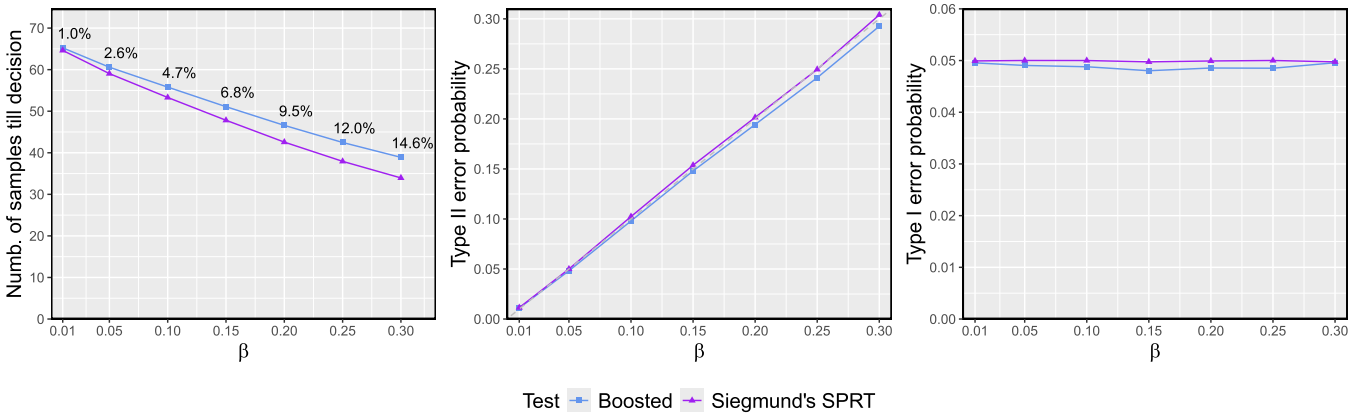


Fig. 6. Comparison of sample size, type II and type I error probability between the SPRT with Siegmund's approximated thresholds and the boosted SPRT in a simple Gaussian testing setup for different desired type II error probabilities (β) and $\alpha = 0.05$. Siegmund's SPRT needs substantially less samples than the boosted SPRT. Siegmund's SPRT controls the error probabilities asymptotically, while the boosted SPRT provides non-asymptotic control.

not apply. For this reason, to the best of our knowledge, our boosted SPRT is the most efficient sequential test for this testing problem. In clinical trials, efficiency is essential for ethical (so that as few patients as possible are treated with ineffective therapies), time-related (effective therapies should be made available to the public as quickly as possible), and economic reasons (recruiting trial participants can be very expensive, especially for rare diseases).

Furthermore, the extensions developed in Section IV offer a range of promising applications. For example, martingale -

based confidence sequences are used for A/B testing in tech companies [43], [44], the sequential sampling without replacement setting appears in risk-limiting election audits [25], [45], [46] and conformal martingales can be used to verify the i.i.d. assumption underlying many statistical and machine learning methods [28], [47]. In all of those settings, improving the employed sequential test by avoiding overshoot with our boosting technique leads to faster and more efficient decision-making.

VII. Summary

We introduced a general approach to uniformly improve nonnegative (super)martingale based sequential tests by avoiding an overshoot at the upper threshold $1/\alpha$ (where α is the desired type I error). We illustrated its application by uniformly improving the conservative power-one SPRT for one-sided null and alternative hypotheses. We also obtained an alternative to the two-sided SPRT with Wald's approximated thresholds, where our alternative provides valid type I and type II error control while often reducing the sample size. Our sequential tests can also be used to tighten confidence sequences and apply in nontrivial settings like sampling without replacement and conformal martingales.

Acknowledgment

The authors are grateful to Wouter M. Koolen for pointing out the possibility of using an inverted SPRT to control the type II error, and to Nick Koning for mentioning the log-optimality of boosting factors. In addition, they would like to thank Yajun Mei, Qunzhi Xu, Venugopal Veeravalli, and Georgios Fellouris for useful discussions. Furthermore, they thank three anonymous reviewers and an associate editor for valuable comments.

Conflicts of Interest

The authors declare that there are no conflicts of interest.

References

[1] A. Wald, "Sequential tests of statistical hypotheses," *Ann. Math. Statist.*, vol. 16, no. 2, pp. 117–186, Jun. 1945.
[2] A. Wald, *Sequential Analysis*. Hoboken, NJ, USA: Wiley, 1947.
[3] A. Wald, "On cumulative sums of random variables," *Ann. Math. Statist.*, vol. 15, no. 3, pp. 283–296, Sep. 1944.
[4] A. Wald and J. Wolfowitz, "Optimum character of the sequential probability ratio test," *Ann. Math. Statist.*, vol. 19, no. 3, pp. 326–339, Sep. 1948.
[5] D. Siegmund, *Sequential Analysis: Tests and Confidence Intervals*. Cham, Switzerland: Springer, 1985.
[6] D. SPiegelhalter, "Risk-adjusted sequential probability ratio tests: Applications to bristol, shipman and adult cardiac surgery," *Int. J. Quality Health Care*, vol. 15, no. 1, pp. 7–13, Feb. 2003.
[7] Y.-C. I. Chang, "Application of sequential probability ratio test to computerized criterion-referenced testing," *Sequential Anal.*, vol. 23, no. 1, pp. 45–61, Dec. 2004.
[8] Y. Chetouani, "A sequential probability ratio test (SPRT) to detect changes and process safety monitoring," *Process Saf. Environ. Protection*, vol. 92, no. 3, pp. 206–214, May 2014.
[9] A. Hapfelmeier, R. Hornung, and B. Haller, "Efficient permutation testing of variable importance measures by the example of random forests," *Comput. Statist. Data Anal.*, vol. 181, May 2023, Art. no. 107689.
[10] M. Steinhilber, M. Schnuerch, and A.-L. Schubert. (2023). *Sprtt: Sequential Probab. Ratio Tests Toolbox*. [Online]. Available: https://CRAN.R-project.org/package=sprtt
[11] S. M. Bottine. (2015). *SPRT: Wald's Sequential Probab. Ratio Test*. [Online]. Available: https://CRAN.R-project.org/package=SPRT
[12] D. A. Darling and H. Robbins, "Some nonparametric sequential tests with power one," *Proc. Nat. Acad. Sci. USA*, vol. 61, no. 3, pp. 804–809, Nov. 1968.
[13] Y. S. Chow, H. Robbins, and D. Siegmund, *Great Expectations: The Theory of Optimal Stopping*. Boston, MA, USA: Houghton Mifflin Company, 1971.
[14] A. Ramdas, J. Ruf, M. Larsson, and W. Koolen, "Admissible anytime-valid sequential inference must rely on nonnegative Martingales," 2020, *arXiv:2009.03167*.
[15] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*, vol. 29. Cham, Switzerland: Springer, 2005.
[16] J. Ville, *Etude Critique De La Notion De Collectif*, 1939.
[17] A. Ramdas, P. Grünwald, V. Vovk, and G. Shafer, "Game-theoretic statistics and safe anytime-valid inference," *Stat. Sci.*, vol. 38, no. 4, pp. 576–601, Nov. 2023.
[18] V. Vovk and R. Wang, "Merging sequential e-values via Martingales," *Electron. J. Statist.*, vol. 18, no. 1, pp. 1185–1205, Jan. 2024.
[19] R. Wang and A. Ramdas, "False discovery rate control with E-values," *J. Roy. Stat. Soc. Ser. B, Stat. Methodology*, vol. 84, no. 3, pp. 822–852, Jul. 2022.
[20] N. W. Koning, "Continuous testing: Unifying tests and E-values," 2024, *arXiv:2409.05654*.
[21] D. Siegmund, "Importance sampling in the Monte Carlo study of sequential tests," *Ann. Statist.*, vol. 4, no. 4, pp. 673–684, Jul. 1976.
[22] H. Robbins, "Statistical methods related to the law of the iterated logarithm," *Ann. Math. Statist.*, vol. 41, no. 5, pp. 1397–1409, Oct. 1970.
[23] H. Robbins and D. Siegmund, "A class of stopping rules for testing parametric hypotheses," in *Proc. 6th Berkeley Symp. Math. Statist. Probab.*, 1985, pp. 293–297.
[24] H. Robbins and D. Siegmund, "The expected sample size of some tests of power one," *Ann. Statist.*, vol. 2, no. 3, pp. 415–436, May 1974.
[25] I. Waudby-Smith, P. B. Stark, and A. Ramdas, "RiLACS: Risk limiting audits via confidence sequences," in *Proc. Electron. Voting: 6th Int. Joint Conf.*, 2021, pp. 124–139.
[26] I. Waudby-Smith and A. Ramdas, "Confidence sequences for sampling without replacement," in *Adv. Neural Inf. Proc. Syst.*, 2020, pp. 20204–20214.
[27] A. Ramdas, J. Ruf, M. Larsson, and W. M. Koolen, "Testing exchangeability: Fork-convexity, supermartingales and e-processes," *Int. J. Approx. Reasoning*, vol. 141, pp. 83–109, Feb. 2022.
[28] V. Vovk, "Testing randomness online," *Stat. Sci.*, vol. 36, no. 4, pp. 595–611, Nov. 2021.
[29] L. Weiss, "Testing one simple hypothesis against another," *Ann. Math. Statist.*, vol. 24, no. 2, pp. 273–281, Jun. 1953.
[30] G. Shafer, "Testing by betting: A strategy for statistical and scientific communication," *J. Roy. Stat. Soc. Ser. A, Statist. Soc.*, vol. 184, no. 2, pp. 407–431, Apr. 2021.
[31] D. Siegmund, "Error probabilities and average sample number of the sequential probability ratio test," *J. Roy. Stat. Soc. Ser. B, Stat. Methodol.*, vol. 37, no. 3, pp. 394–401, Jul. 1975.
[32] M. Woodroofe, "A renewal theorem for curved boundaries and moments of first passage times," *Ann. Probab.*, vol. 4, no. 1, pp. 67–80, Feb. 1976.
[33] T. L. Lai and D. Siegmund, "A nonlinear renewal theory with applications to sequential analysis I," *Ann. Statist.*, vol. 5, no. 5, pp. 946–954, Sep. 1977.
[34] K. W. Kemp, "Formulae for calculating the operating characteristic and the average sample number of some sequential tests," *J. Roy. Stat. Soc. Ser. B, Stat. Methodol.*, vol. 20, no. 2, pp. 379–386, Jul. 1958.
[35] E. S. Page, "An improvement to Wald's approximation for some properties of sequential tests," *J. Roy. Stat. Soc. Ser. B, Stat. Methodol.*, vol. 16, no. 1, pp. 136–139, Jan. 1954.
[36] A. Tartakovsky, I. Nikiforov, and M. Basseville, *Sequential Analysis: Hypothesis Testing and Changepoint Detection*. Boca Raton, FL, USA: CRC Press, 2014.
[37] P. Armitage, C. K. McPherson, and B. C. Rowe, "Repeated significance tests on accumulating data," *J. Roy. Stat. Soc., Ser. A (General)*, vol. 132, no. 2, pp. 235–244, 1969.
[38] C. Jennison and B. W. Turnbull, *Group Sequential Methods With Applications to Clinical Trials*. Boca Raton, FL, USA: CRC Press, 1999.

[39] G. Wassmer and W. Brannath, *Group Sequential and Confirmatory Adaptive Designs in Clinical Trials*, vol. 301. Cham, Switzerland: Springer, 2016.
[40] P. Grünwald, R. D. Heide, and W. M. Koolen, "Safe testing," *J. Roy. Stat. Soc. Ser. B, Stat. Methodol.*, vol. 86, no. 5, pp. 1091–1128, 2024.
[41] V. Vovk and R. Wang, "E-values: Calibration, combination and applications," *Ann. Statist.*, vol. 49, no. 3, pp. 1736–1754, Jun. 2021.
[42] Z. Xu, A. Solari, L. Fischer, R. de Heide, A. Ramdas, and J. Goeman, "Bringing closure to false discovery rate control: A general principle for multiple testing," 2025, *arXiv:2509.02517*.
[43] R. Johari, P. Koomen, L. Pekelis, and D. Walsh, "Always valid inference: Continuous monitoring of A/B tests," *Operations Res.*, vol. 70, no. 3, pp. 1806–1821, May 2022.
[44] A. Maharaj et al., "Anytime-valid confidence sequences in an enterprise A/B testing platform," in *Proc. Companion ACM Web Conf.*, Apr. 2023, pp. 396–400.
[45] M. Lindeman, P. B. Stark, and V. S. Yates, "BRAVO: Ballot-polling risk-limiting audits to verify outcomes," *EVT/WOTE*, vol. 125, p. 129, Oct. 2012.
[46] P. B. Stark, "ALPHA: Audit that learns from previously hand-audited ballots," *Ann. Appl. Statist.*, vol. 17, no. 1, pp. 641–679, Mar. 2023.
[47] V. Vovk, A. Gammerman, and C. Saunders, "Machine-learning applications of algorithmic randomness," in *Proc. 16th Int. Conf. Mach. Learn.*, 1999, pp. 444–453.

Lasse Fischer received the B.Sc., M.Sc., and the Ph.D. degrees in mathematics from the University of Bremen, Bremen, Germany, in 2019, 2022, and 2025, respectively. From 2022 to 2025, he was a Research Assistant with the University of Bremen. During this time, he spent four months on a research visit with Carnegie Mellon University, Pittsburgh, USA. He is currently a Post-Doctoral Researcher with the University of Bremen. His main research interests include (online) multiple testing, anytime-valid sequential inference, and post-hoc decision-making.

Aaditya Ramdas received the Ph.D. degree in statistics and machine learning in 2015. He is currently a tenured Associate Professor with the Department of Statistics and Data Science and the Machine Learning Department, Carnegie Mellon University. He works on mathematical statistics and learning. He is best known for his work on e-values, confidence sequences, multiple testing, and conformal prediction. He is a fellow of IMS. He received the Bernoulli New Researcher Award, the NSF CAREER Award, the IIT Bombay Young Alumnus Achievement Award, and the Sloan Fellowship. He was the Program Chair of AISTATS.